

What is Computer?

A **computer** is a machine that manipulates data according to a list of instructions.

A **computer** is a device capable of performing computations and making logical decisions at speed millions and even billion of times faster than human beings can.

Computer is an electronic machine which is used for data processing. The output which comes after processing data through computer is known as Information.



Why Computers?



History of Computers:

- ➔ Zero th generation (1642 to 1945)
- ➔ First Generation (1945 to 1955)
- ➔ Second generation (1955 to 1965)
- ➔ Third generation (1966 to 1970)
- ➔ Fourth generation (1971 to 1980)
- ➔ Fifth generation (1980 to present)

Zero th generation (1642 to 1945)

➔ Mechanical calculators was developed as computing device.



Courtesy IBM (www.ibm.com)

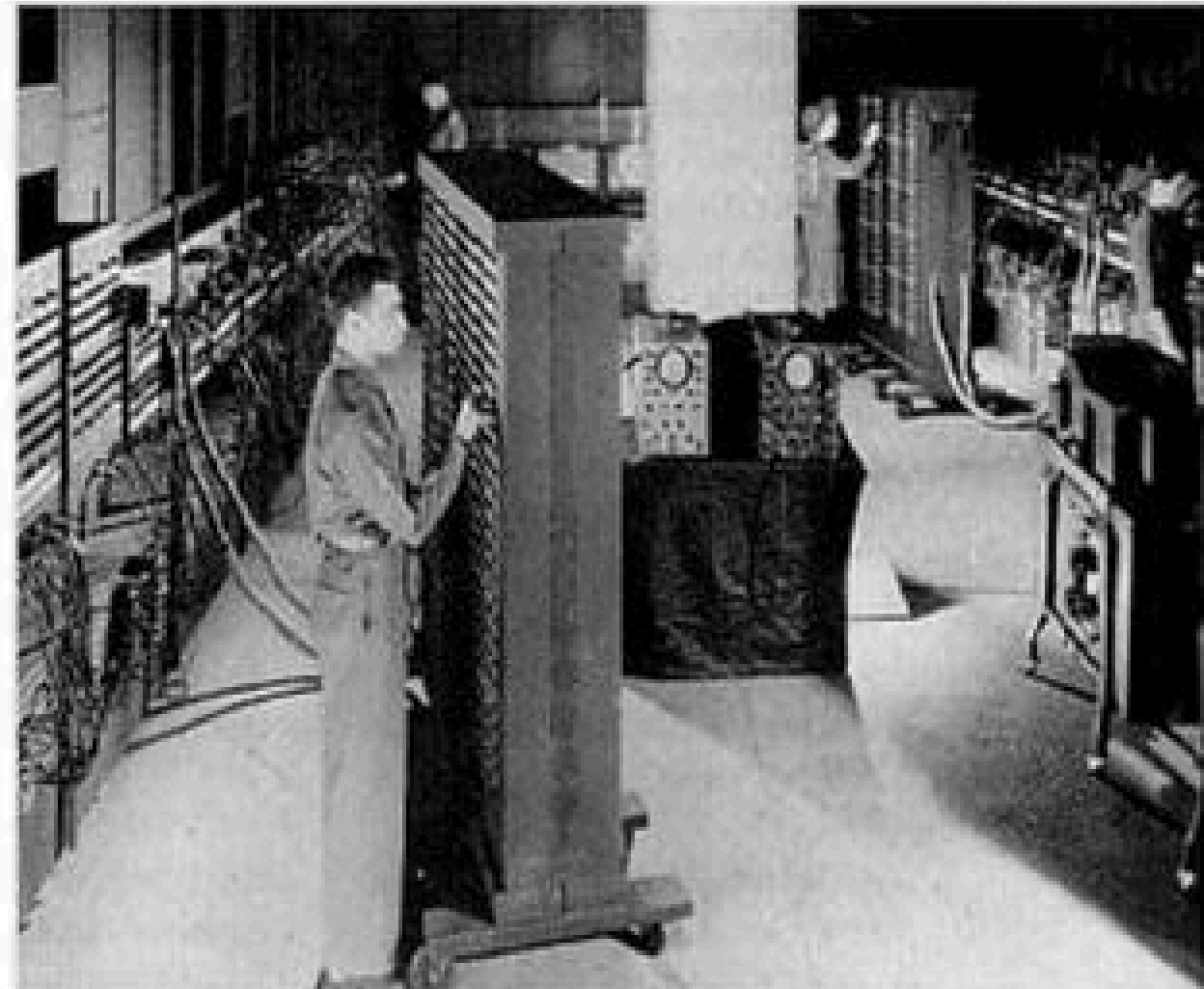
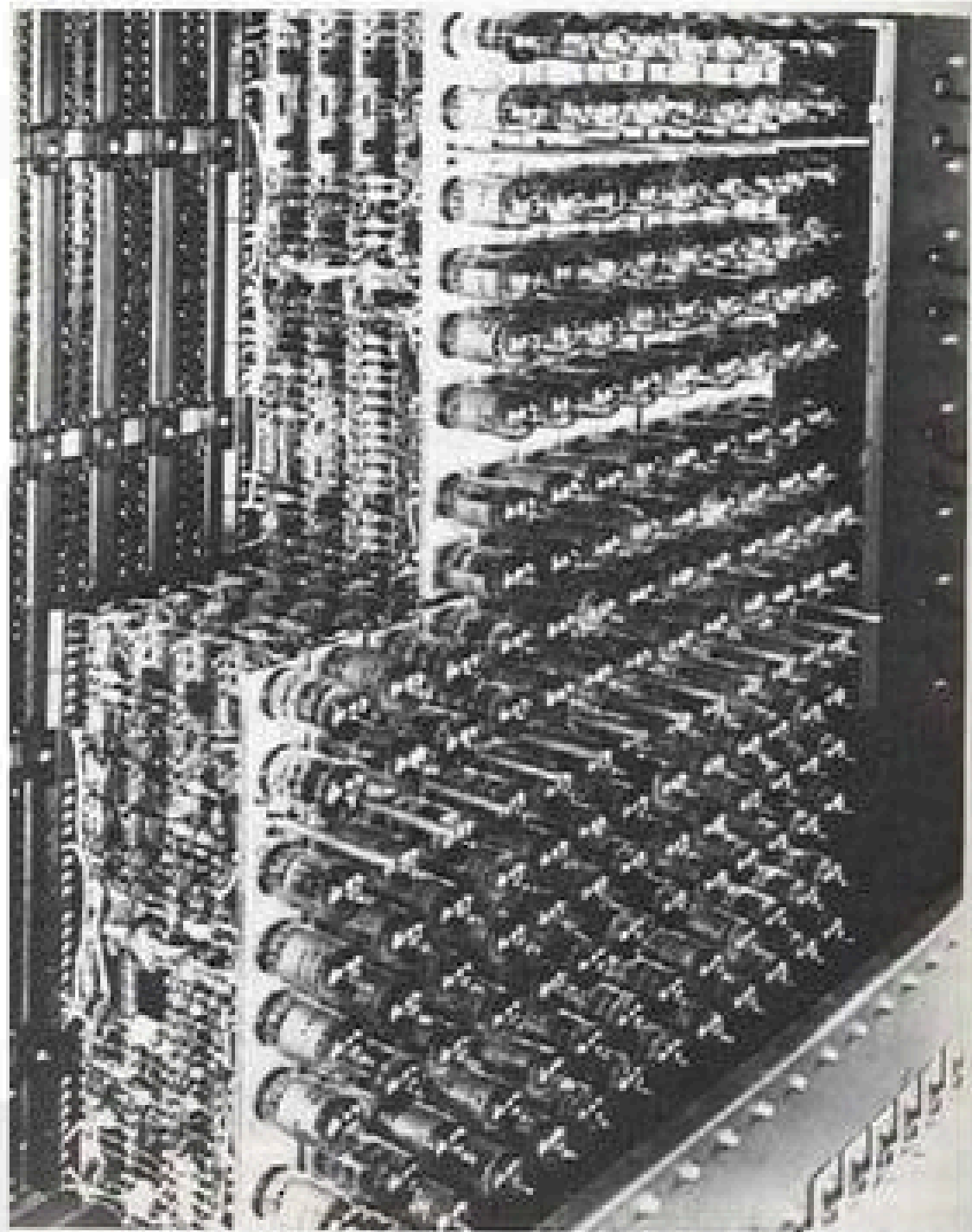


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First Generation (1945 to 1955)

- ➔ In 1946 first electronic computer ENAIC (Electronic Numerical Integrator and Calculator) was developed by Mr. J.Presper Eckert and Mr. John Manuchly at University of Pennsylvania.
- ➔ It uses 18,000 vacuum tubes and its cost was \$500,000.
- ➔ Its weight was 30 tons and occupied a 30 by 50 foot space.
- ➔ It produced large amount of heat.
- ➔ Its speed was very slow.

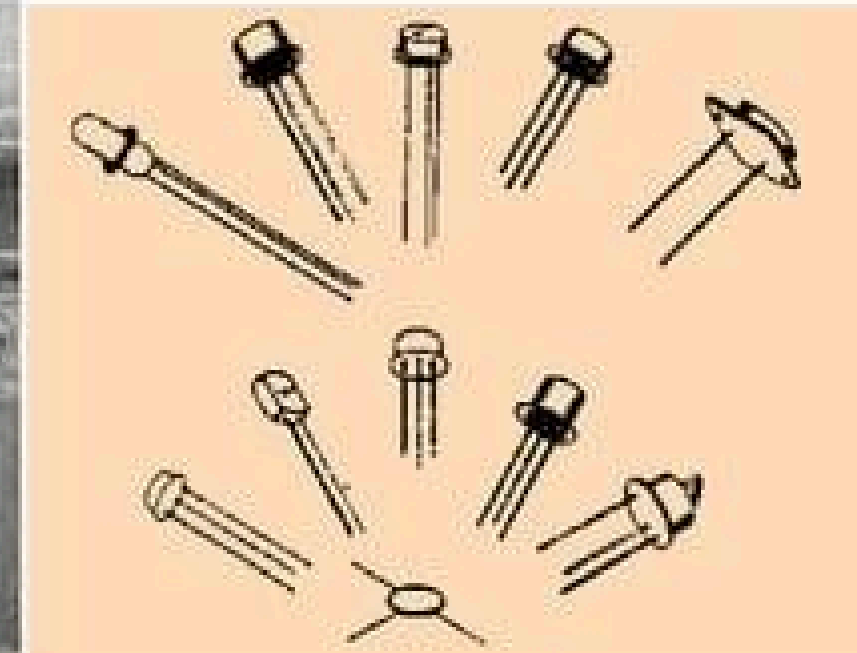
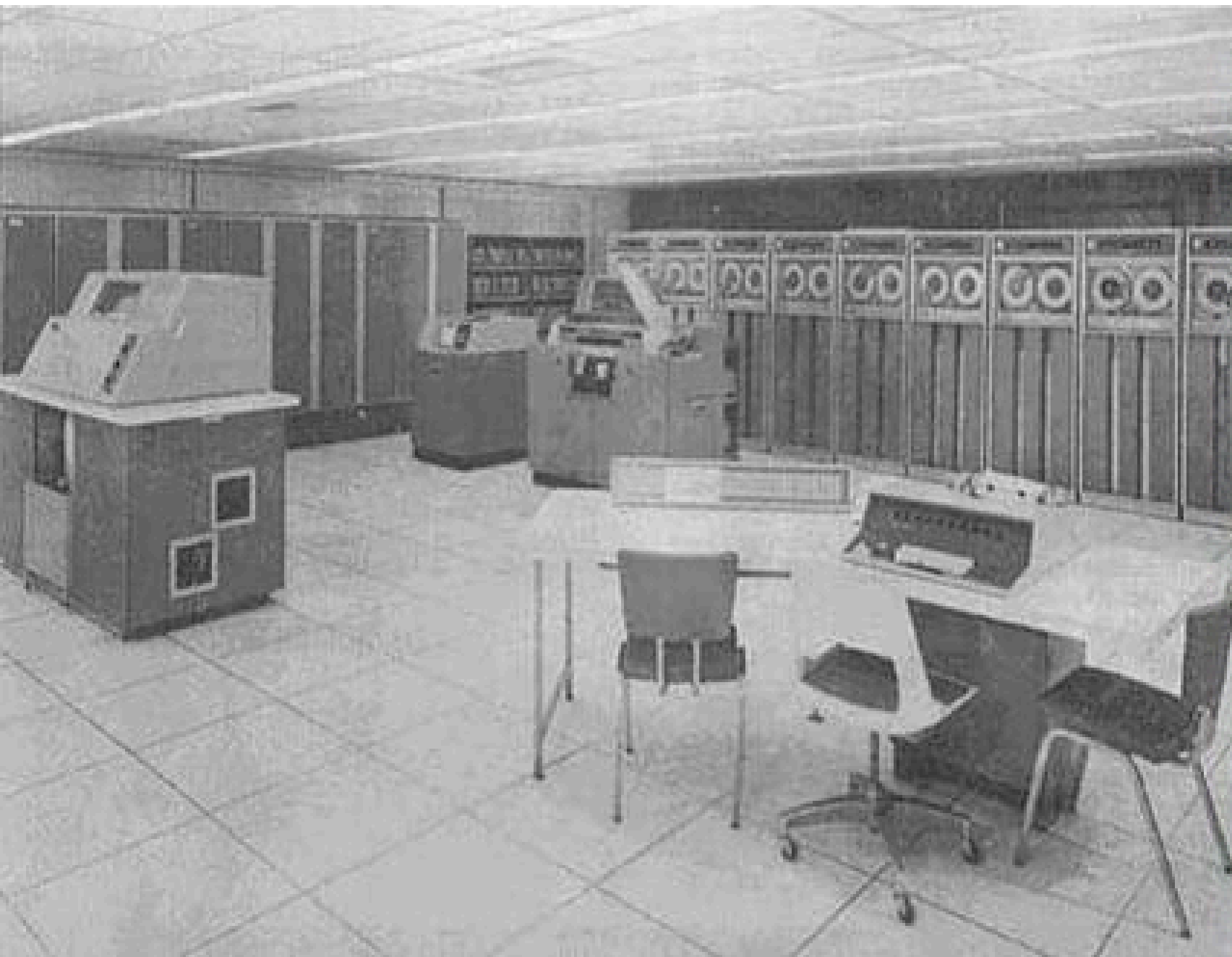
First Generation (1945 to 1955)



Second generation (1955 to 1965)

- ➔ In this second generation vacuum tubes was replaced by transistors.
- ➔ Speed / performance of this generation computers was improved in comparison to first generation computers.
- ➔ Size, price , and heat produced reduced by using transistors.

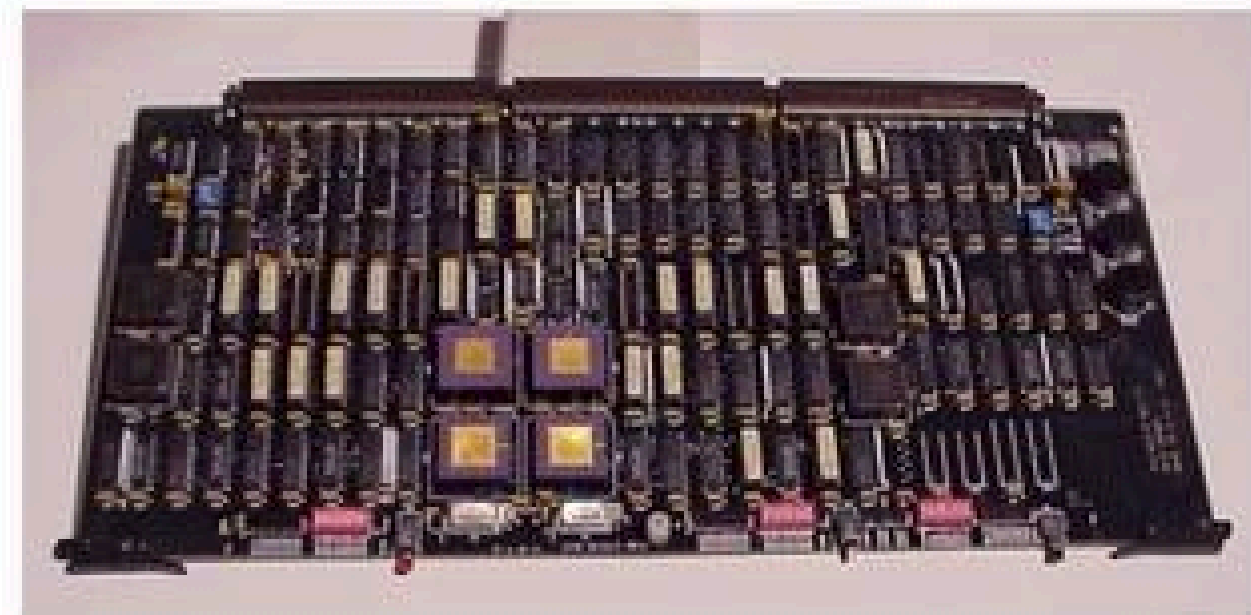
Second generation (1955 to 1965)



Third generation (1966 to 1970)

- ➔ In this third generation transistors was replaced by IC's (Integrated Circuits i.e fabrication of thousands electronic components on single silicon chip).
- ➔ Using this drastic reduction in the size of computers.
- ➔ Speed / performance was high in comparison to third generation.
- ➔ Heat produced by computers was reduced.

Third generation (1966 to 1970)



Fourth generation (1971 to 1980)

- ➔ In this generation IC's are replaced by LSI (Large Scale Integrated Circuits : more number of electronic components on silicon chip)
- ➔ These are low cost, small size and high performance in comparisons to third generation computers

Fifth generation (1980 to present)

- In this generation LSI are replaced by VLSI (Very Large Scale Integrated Circuits)
- Size and cost of these computers is very less and performance is very high



A computer can be divided into two main categories:

1. Hardware

Hardware is basically anything that you can touch with your fingers.

2. Software

For computer hardware to work it must follow a set of instructions that is supplied to it as software.

Hardware

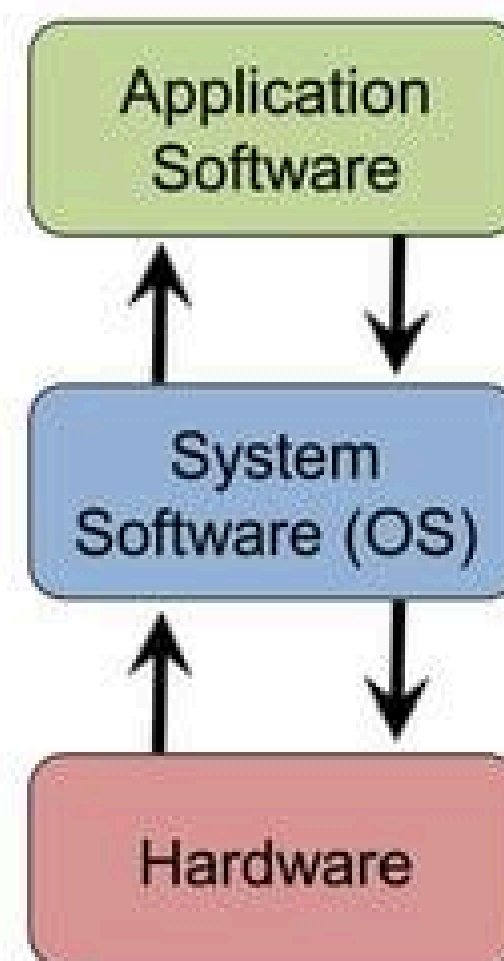
Hardware refers to the computer's physical components,

- Computer Case
- CPU (central processing unit...Pentium chip)
- Monitor
- Keyboard & Mouse
- Disk Drive, Zip Drive, CD-ROM, DVD,
- Hard Drive
- Memory (RAM)
- Speakers
- Printer

Software

The term software refers to the set of instructions that directs the hardware to accomplish a task.

1. System Software
2. Application Software
3. Utility Software



1. System Software/Operating System

System Software are used to control all components of computer and serves as an interface between user and hardware.

1. Control all Components of Computer
2. Interface b/w user and Hardware

2. Application Software

Application software is used or design for the specific purpose of the user.

for example

MS-WORD

MS-EXCEL

Corel Draw

Game

MS-POWER POINT

Acdsee

Paint

MS-ACCESS

3. Utility Software

Utility Software is used to remove any problem or solve a complex situation in computer.

for example

Partition Magic

Antivirus

Security Software

Backup utility

Data Recovery

Win Ghost

Bits n Bytes

→ Bit:

One digit, either **0** or **1**

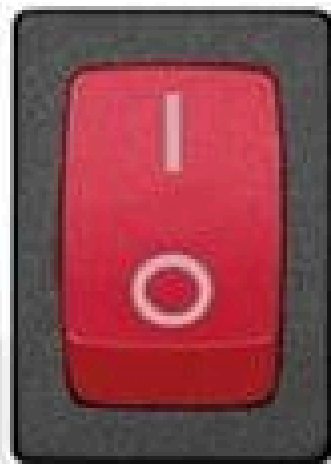
→ Byte:

any **combination** of 0 or 1.

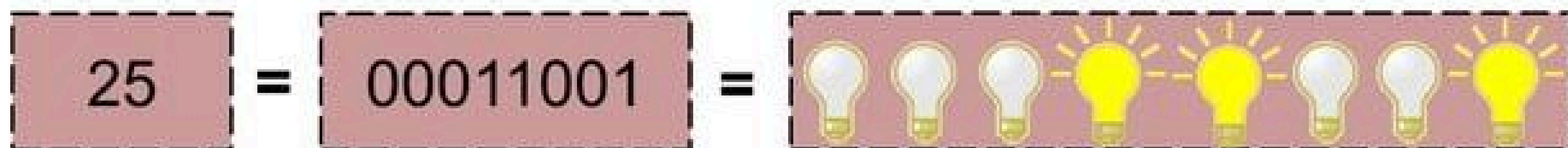
also called an **octet**.

Computer only understand 0 or 1

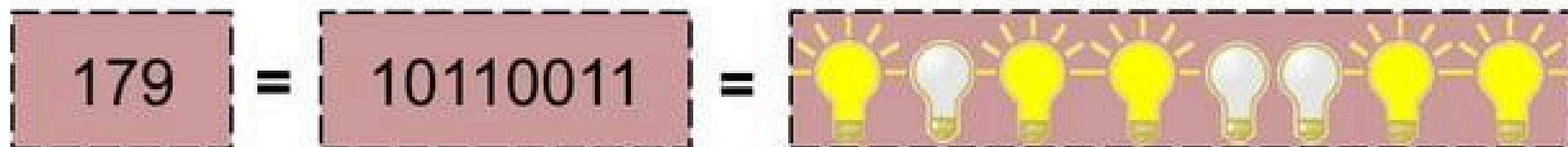
0 → OFF
1 → ON



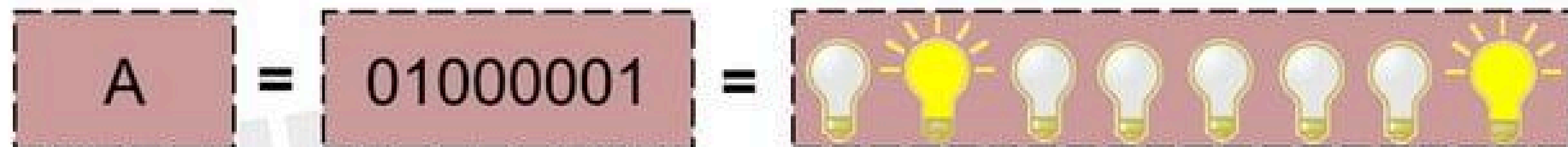
The number **25** stored as **8** bits using the binary number system:



The number **179** stored as **8** bits using the binary number system:



The letter **A** stored as **8** bits using ASCII code:



➔ **INPUT**

Information for the Exam

➔ **PROCESSING**

➔ **OUTPUT**

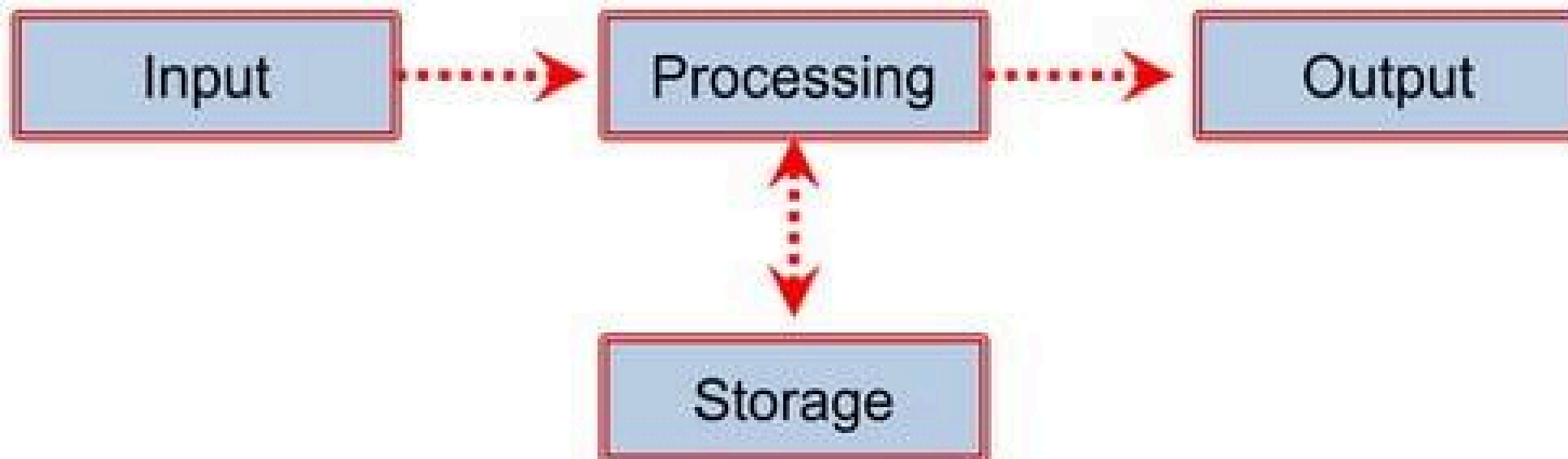


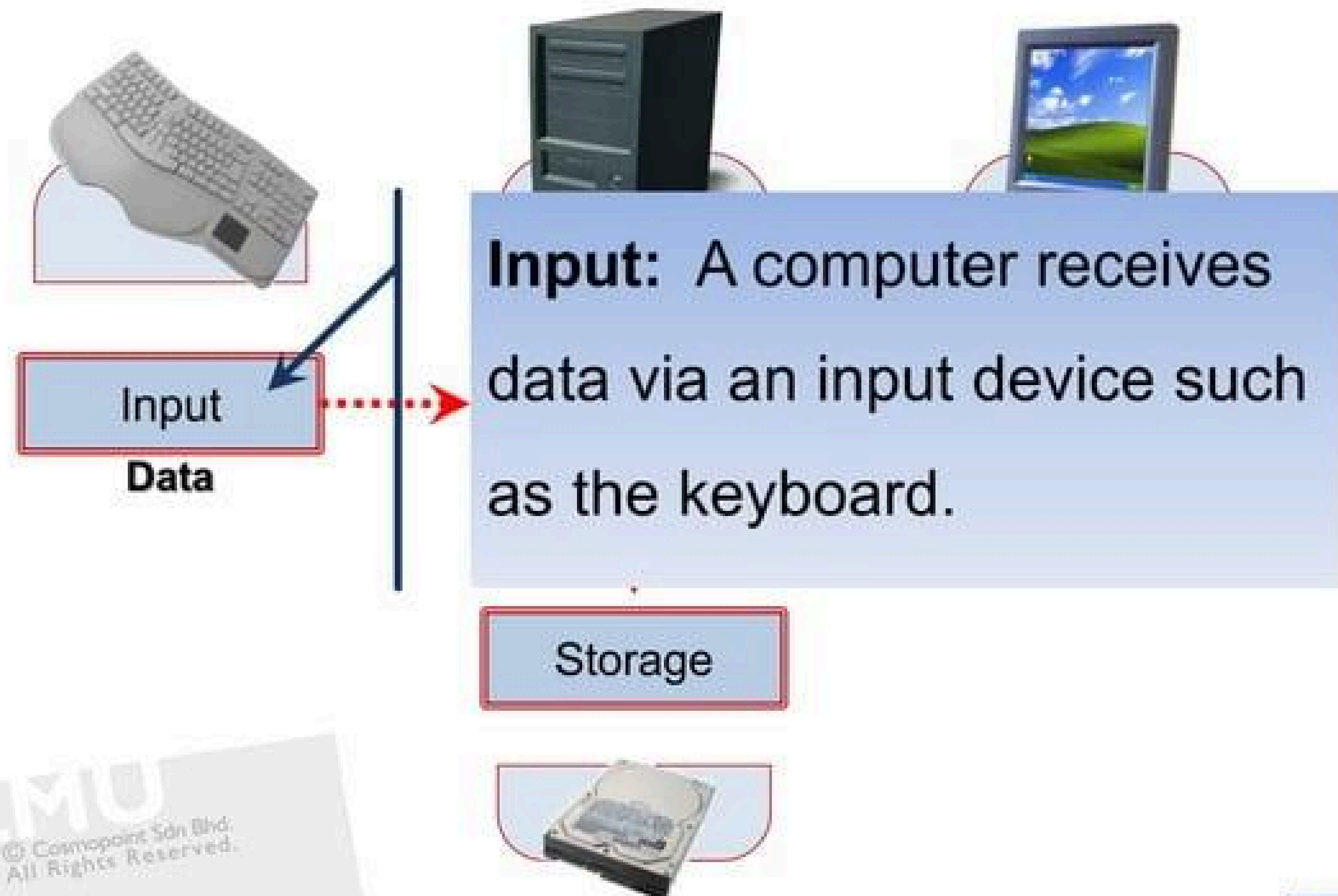
Exam Results!

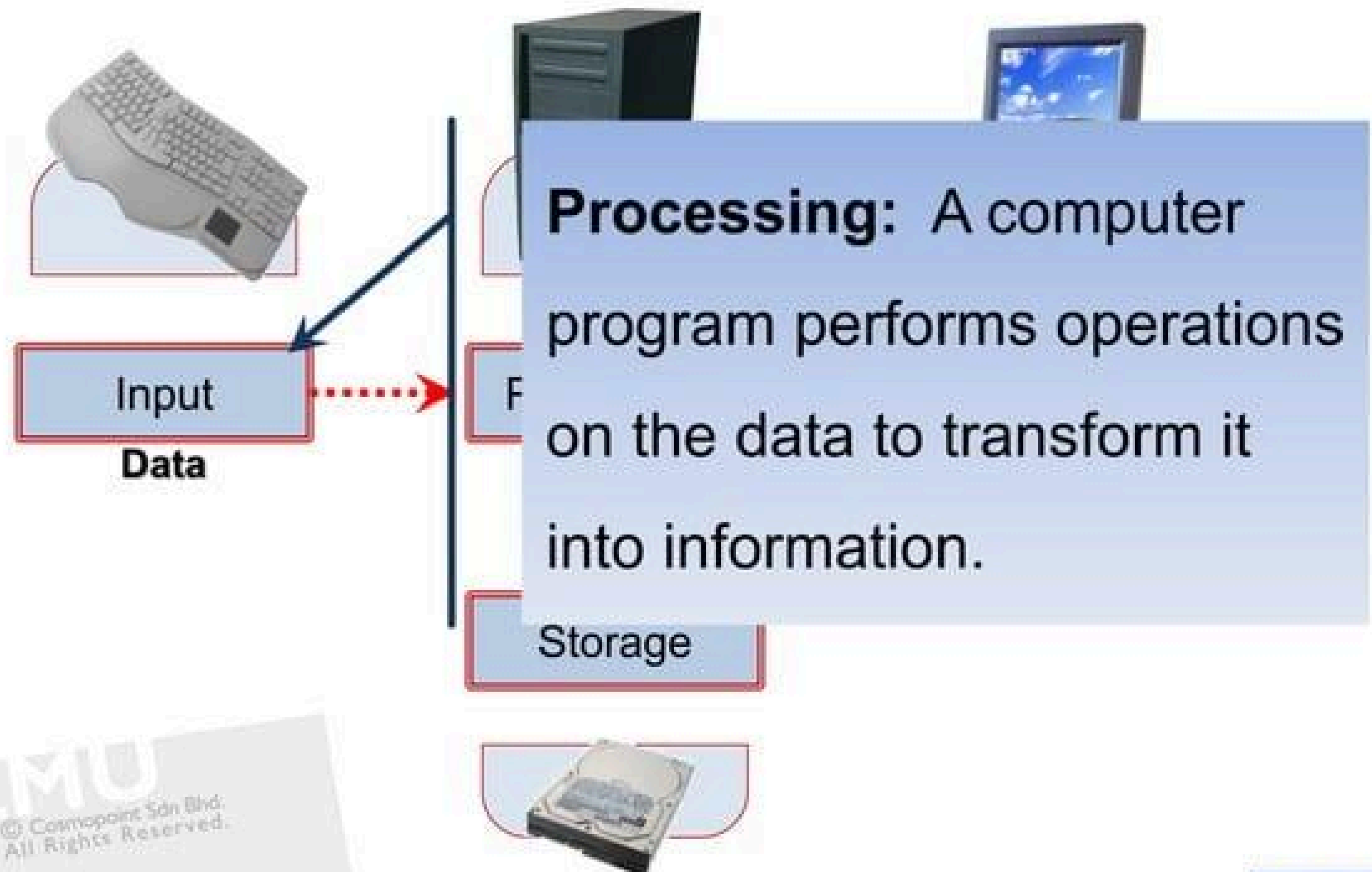
Information Processing Cycle

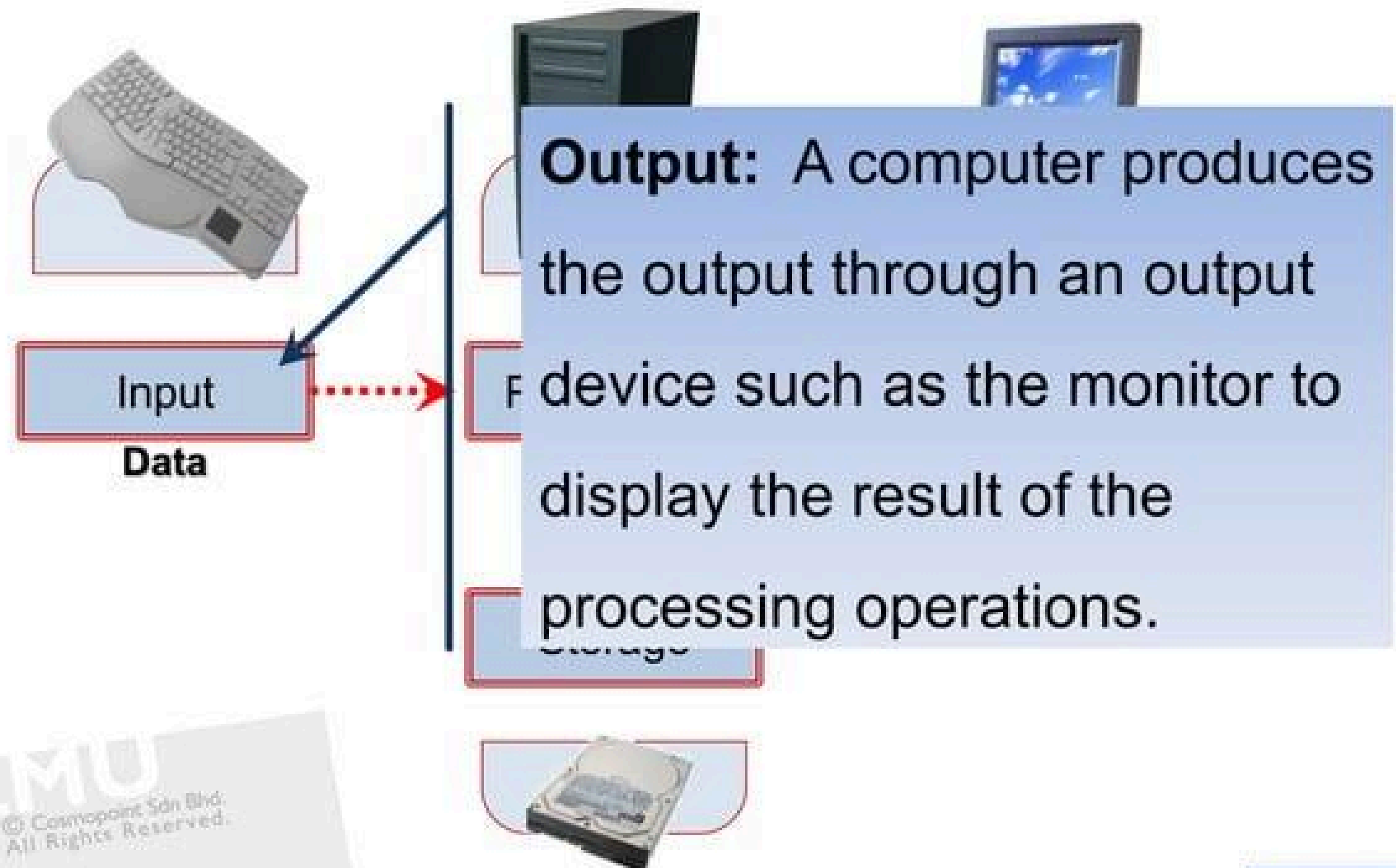
Consists of four parts where each part involves one or more specific components of the computer.

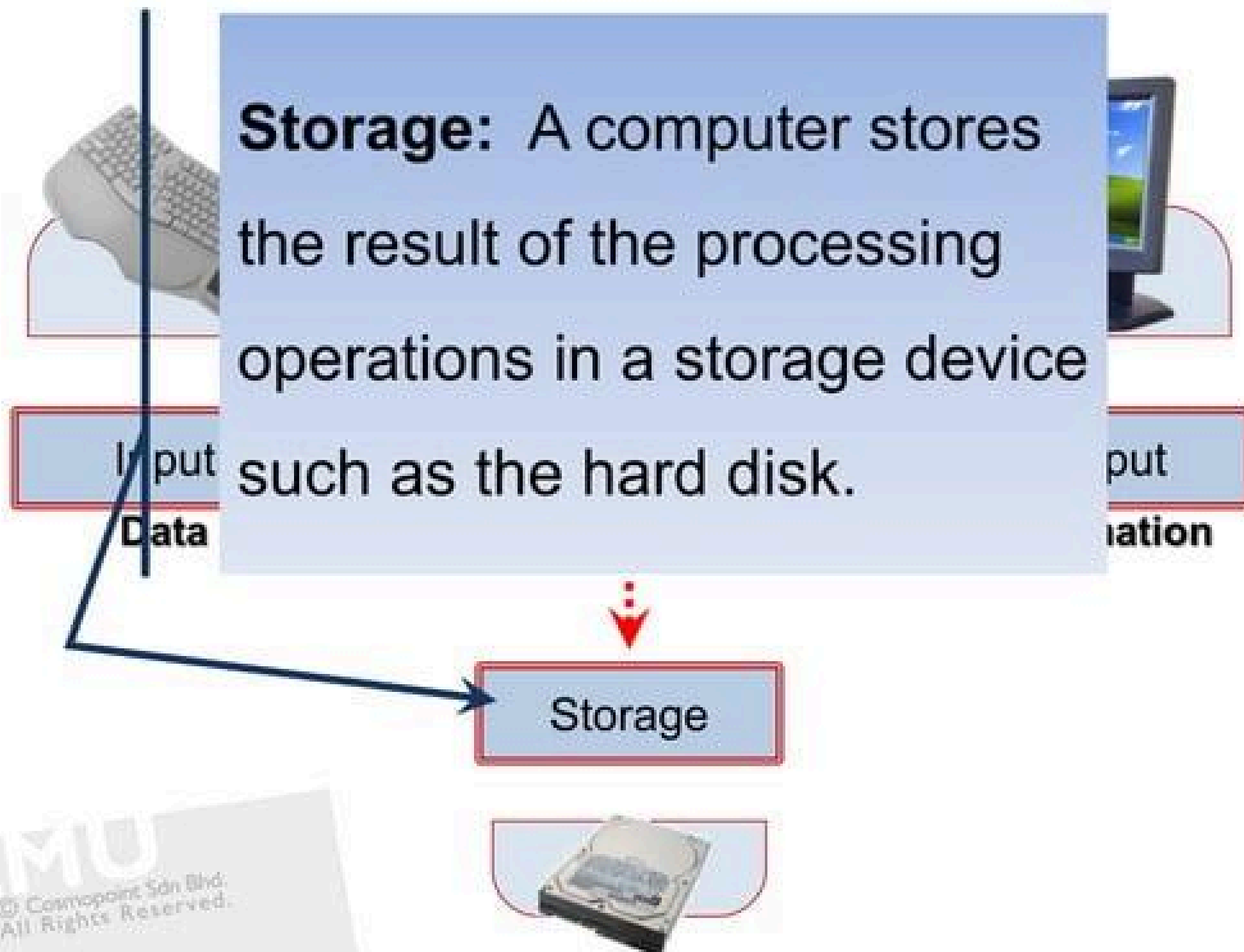
1. Input
2. Processing
3. Output
4. Storage

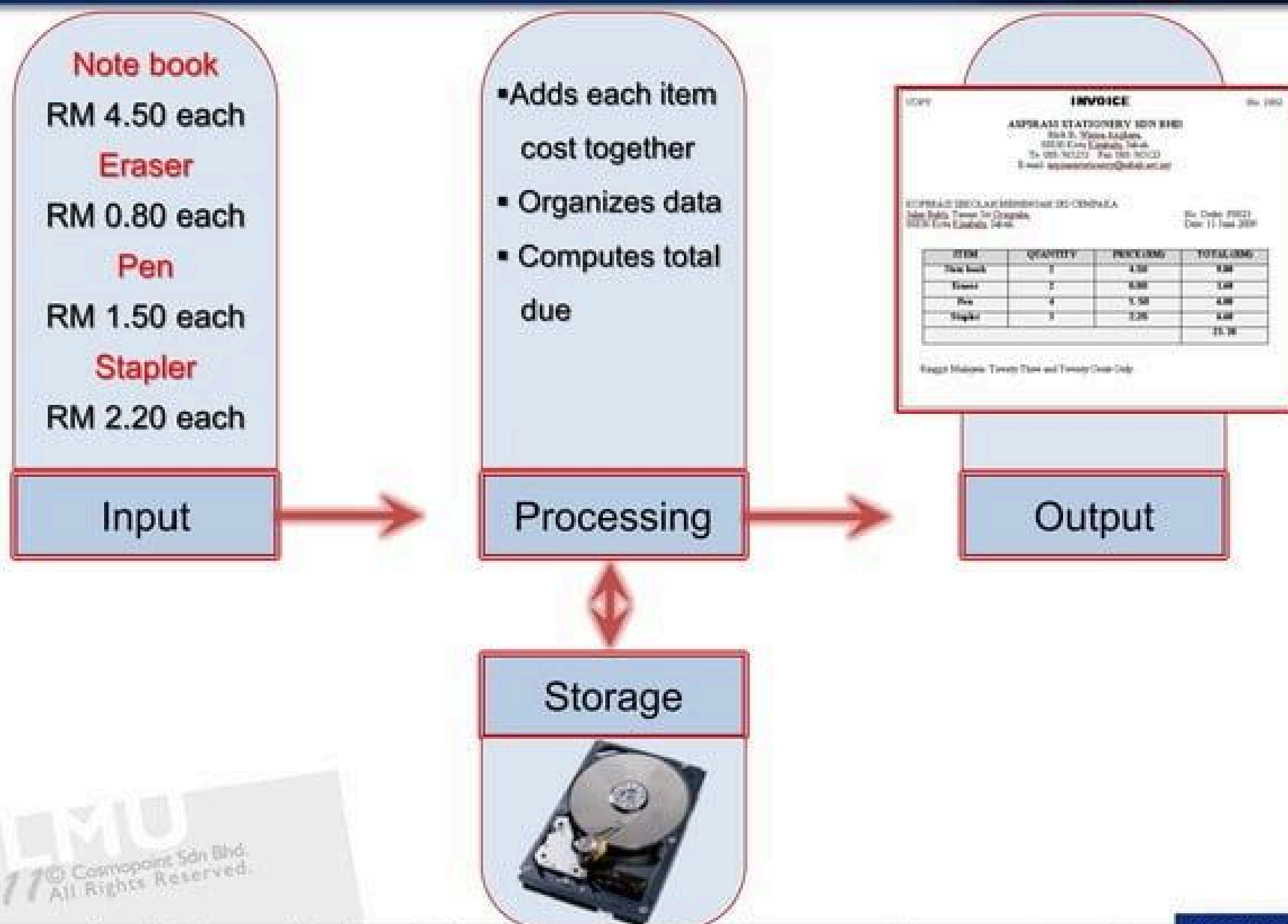












So

Computer is an Electronic device which can:

1. accept data
2. process it
3. give the result after that processing
4. and also store processed data



Storage

PC four major functions are:

- ➔ Input data
- ➔ Process data
- ➔ Output information
- ➔ Store data and information



Data vs. Information

Data is a representation of a fact or idea

Number

examples of data

Word

3547 Ahmad Kuala Lumpur Malaysia

Picture

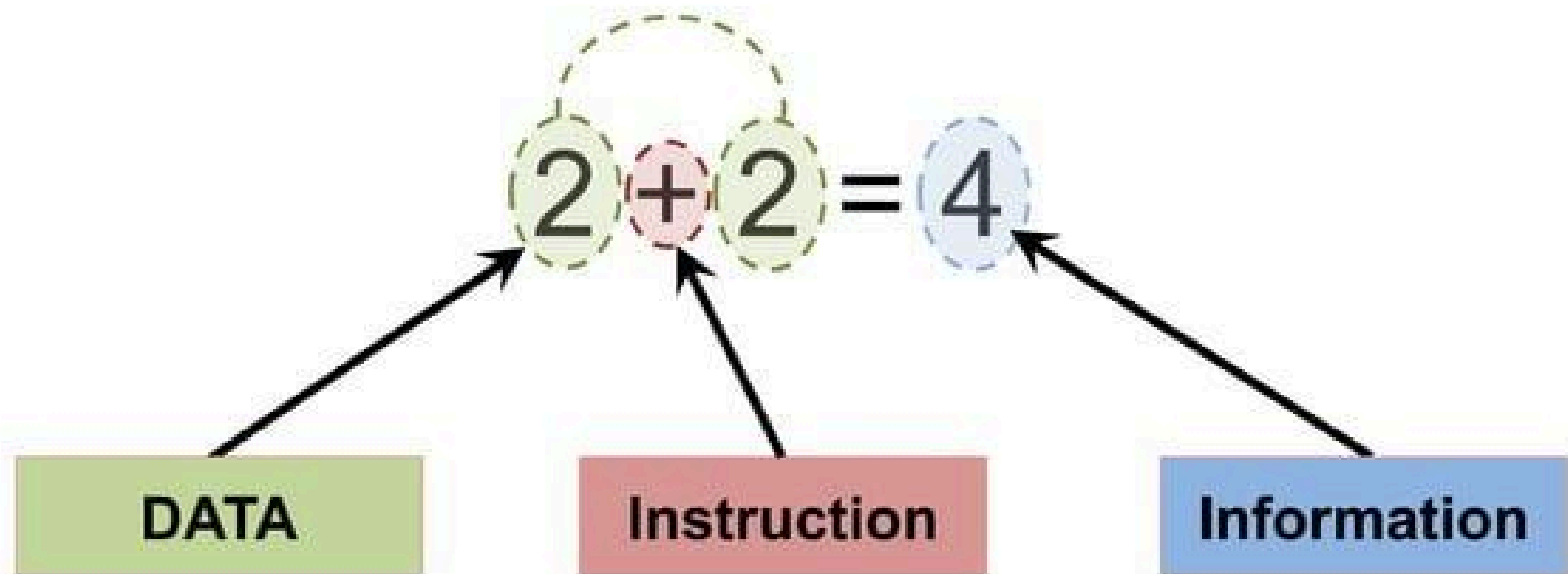
examples of information

Sound

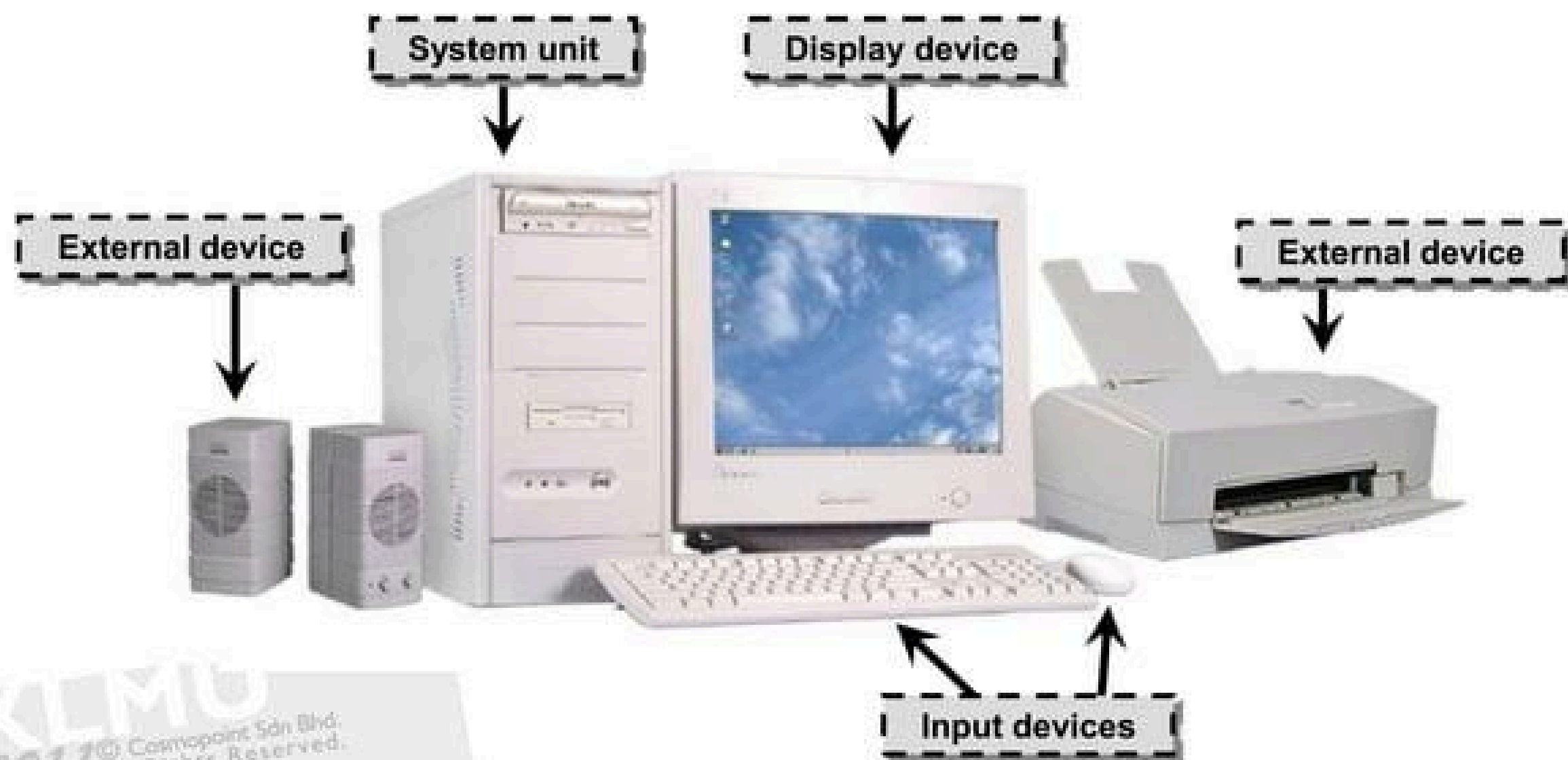
Roll No. 3547 Name-Ahmad City-Kuala Lumpur
Country-Malaysia

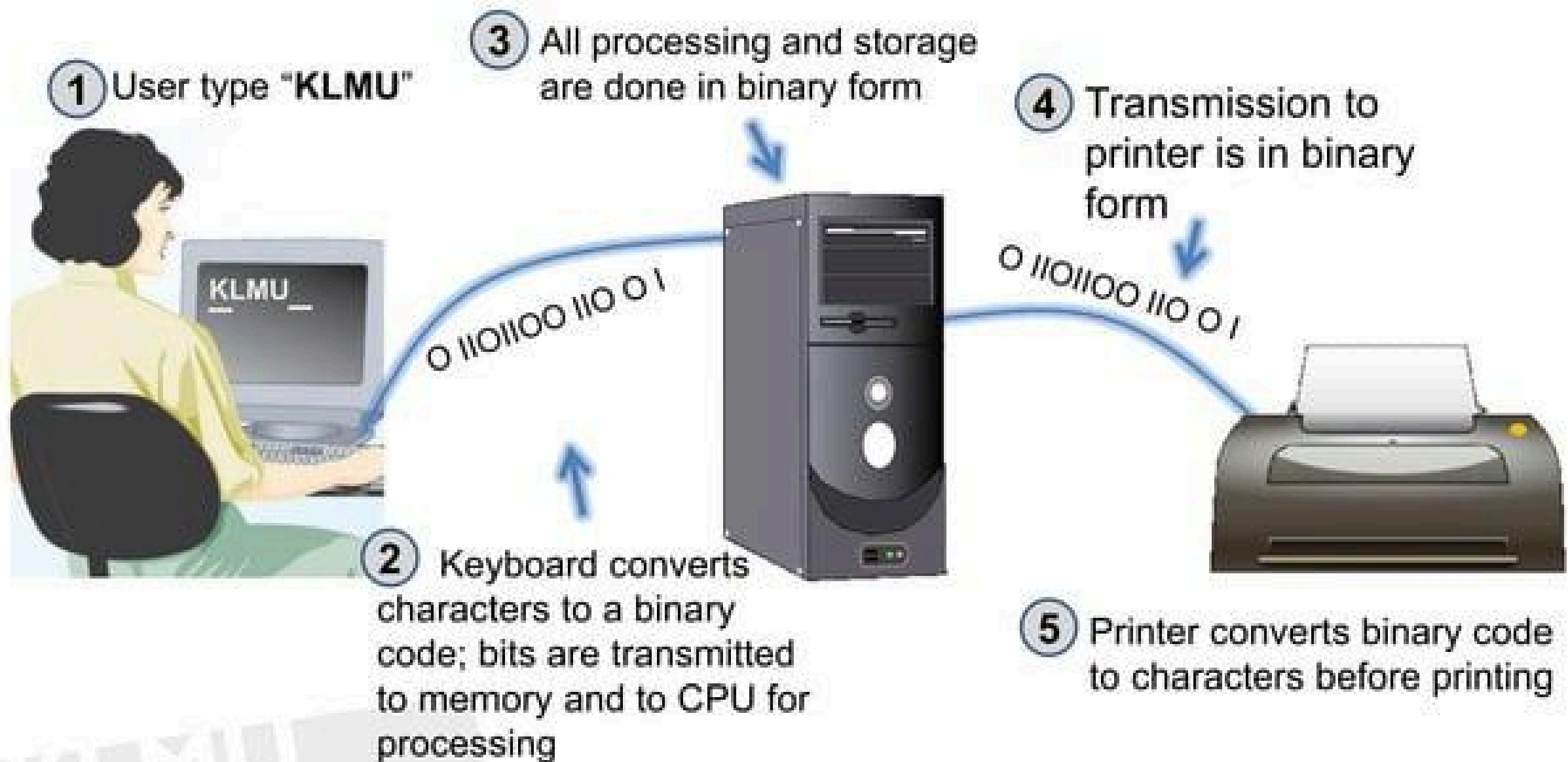
Information is data that has been organized or presented in a meaningful.

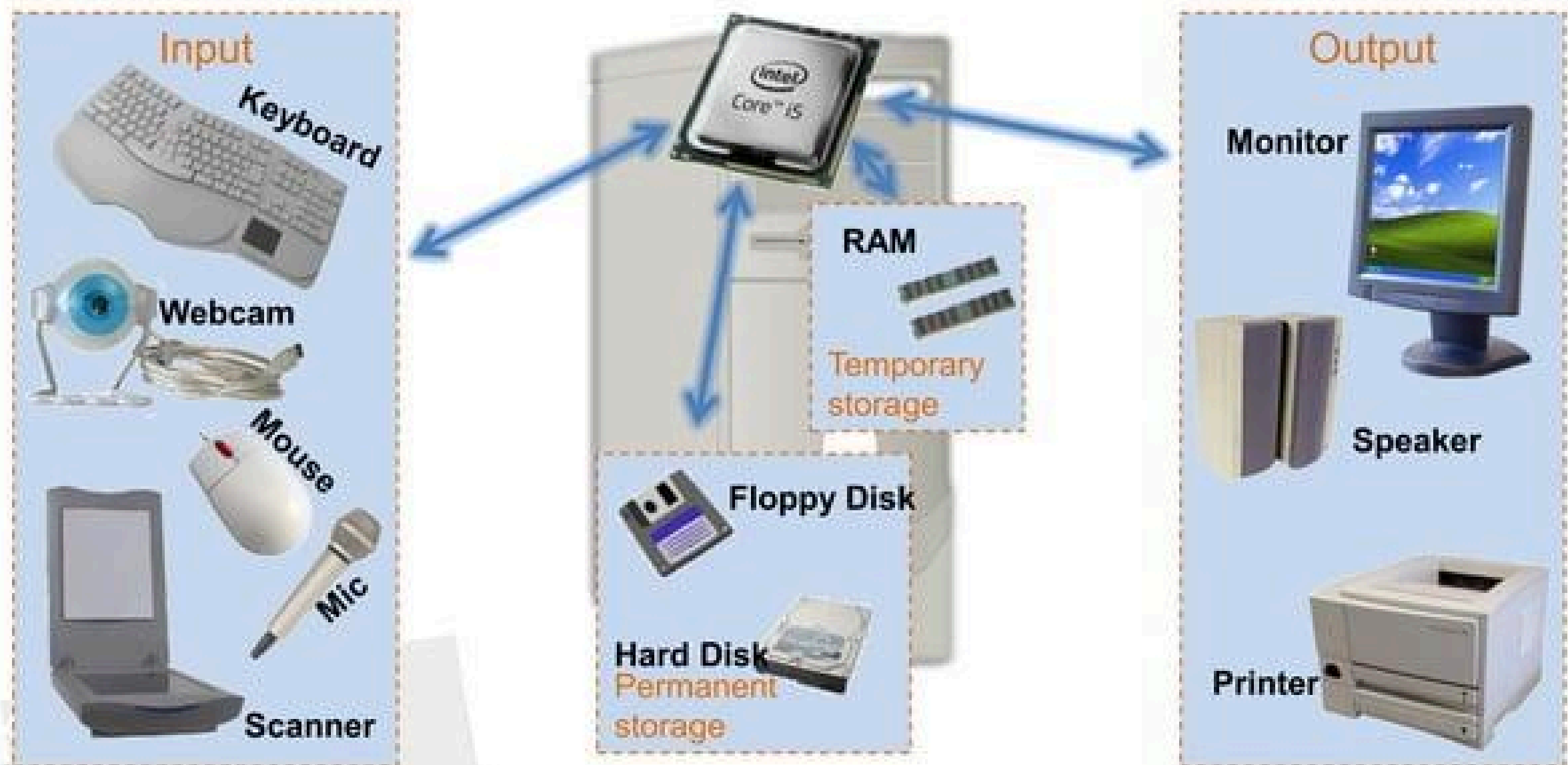
Data vs. Instruction



A Personal Computer:







Enter data to be processed

- Keyboard
- Scanners
- Mouse
- Trackball
- Touch screen
- Microphone
- Game Controller
- Digital camera



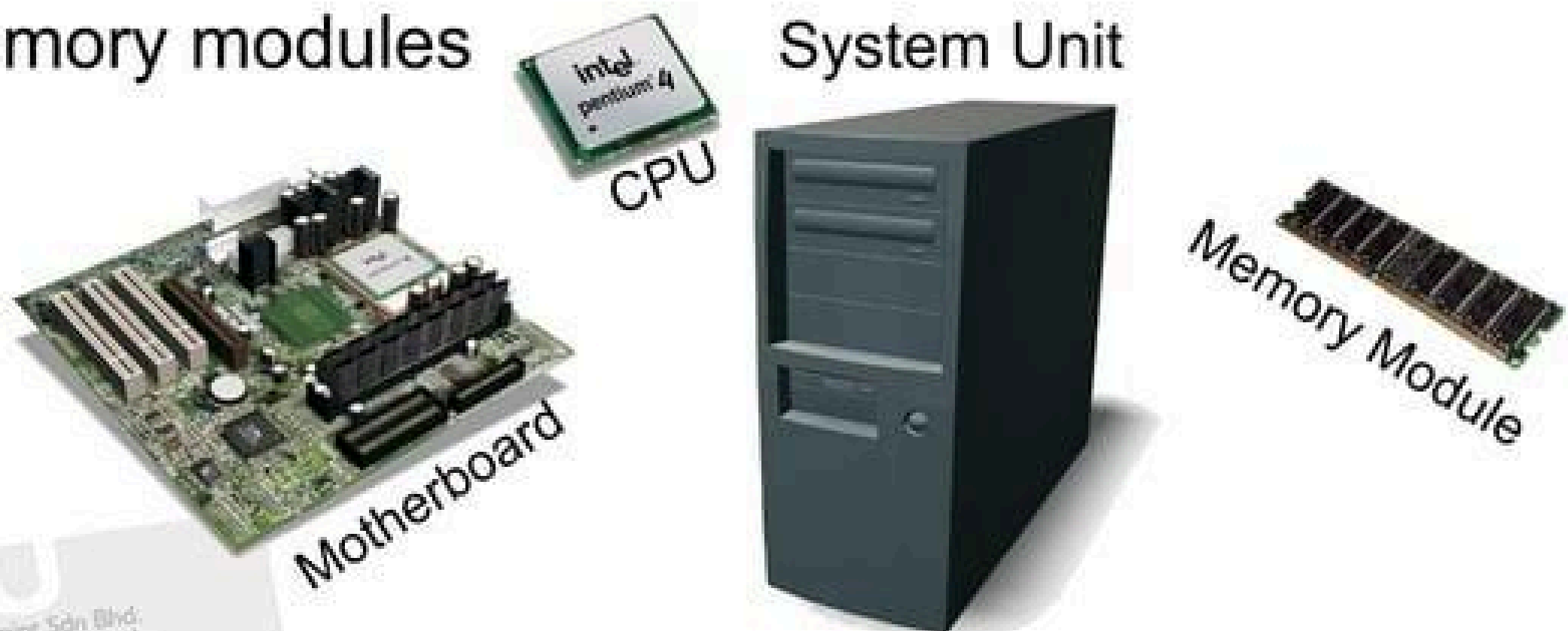
System Unit

Cabinet that houses all components

➔ Motherboard

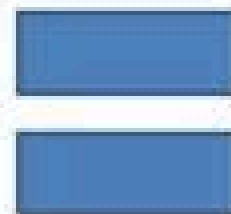
➔ CPU

➔ Memory modules

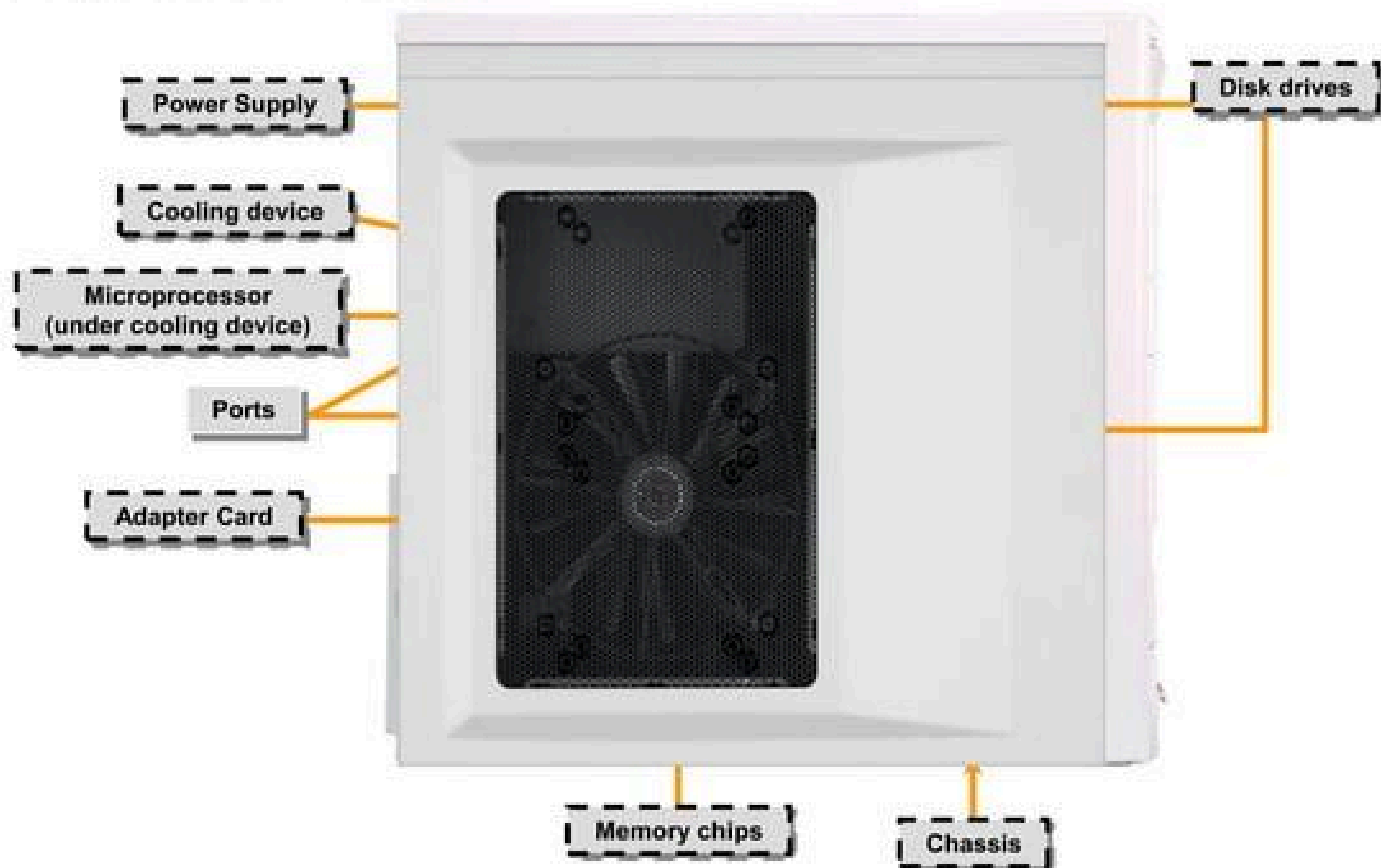


A processor present in **system unit** is the part that is like a computer's brain.

A **system unit** is a personal computer component that houses other devices necessary for the computer to function.



The System Unit



Display Device

A ***display device*** is a personal computer component that enables users to view the text and graphical data associated with a computer program.



Input Devices

An ***input device*** is a personal computer component that enables users to enter data or instructions into a computer.



External Devices

There are several categories of external devices:

- ➔ Microphone
- ➔ Digital camera
- ➔ Scanner
- ➔ Speakers
- ➔ Printer
- ➔ Network device
- ➔ External drive



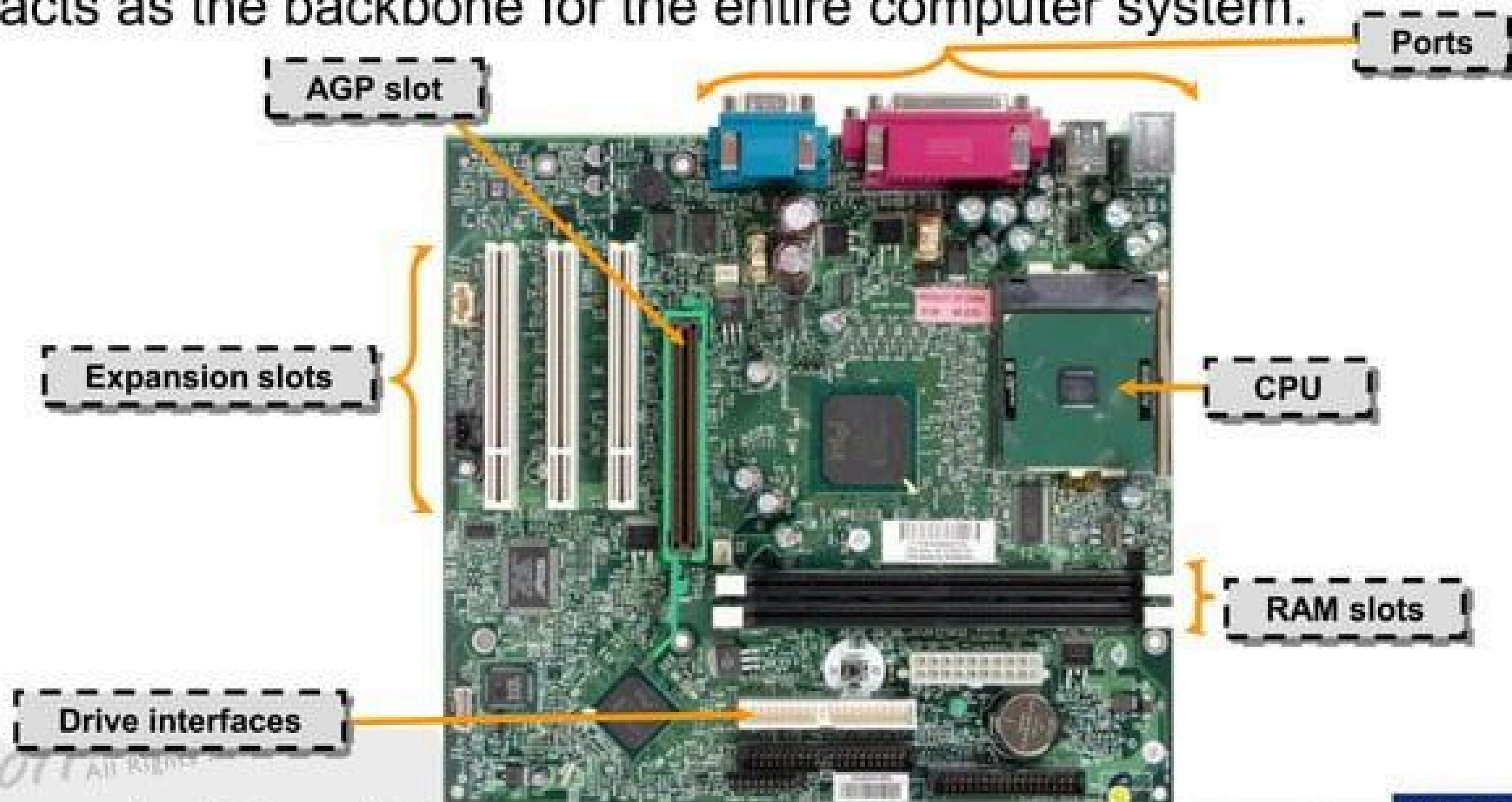
System Unit Components

The system unit itself has several important sub-components, such as:

- ➔ System Board
- ➔ Central Processing Unit (CPU)
- ➔ Memory
- ➔ System Bus
- ➔ Storage devices

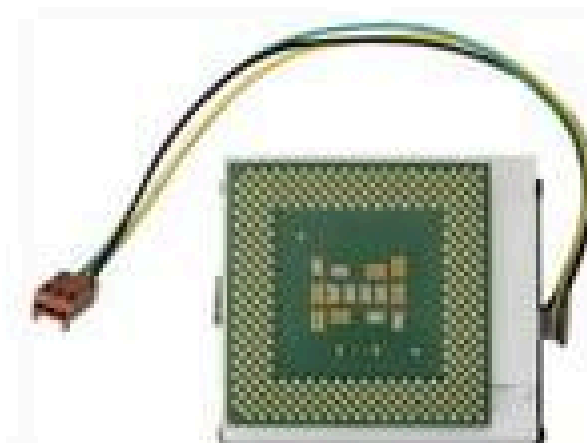
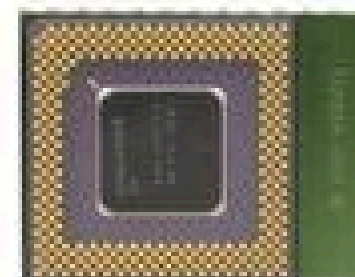
The System Board

The *system board* is the personal computer component that acts as the backbone for the entire computer system.



The Central Processing Unit (CPU)

The *central processing unit (CPU)*, sometimes called microprocessor or just processor, is the real brains of the computer and is where most of the calculations take place.



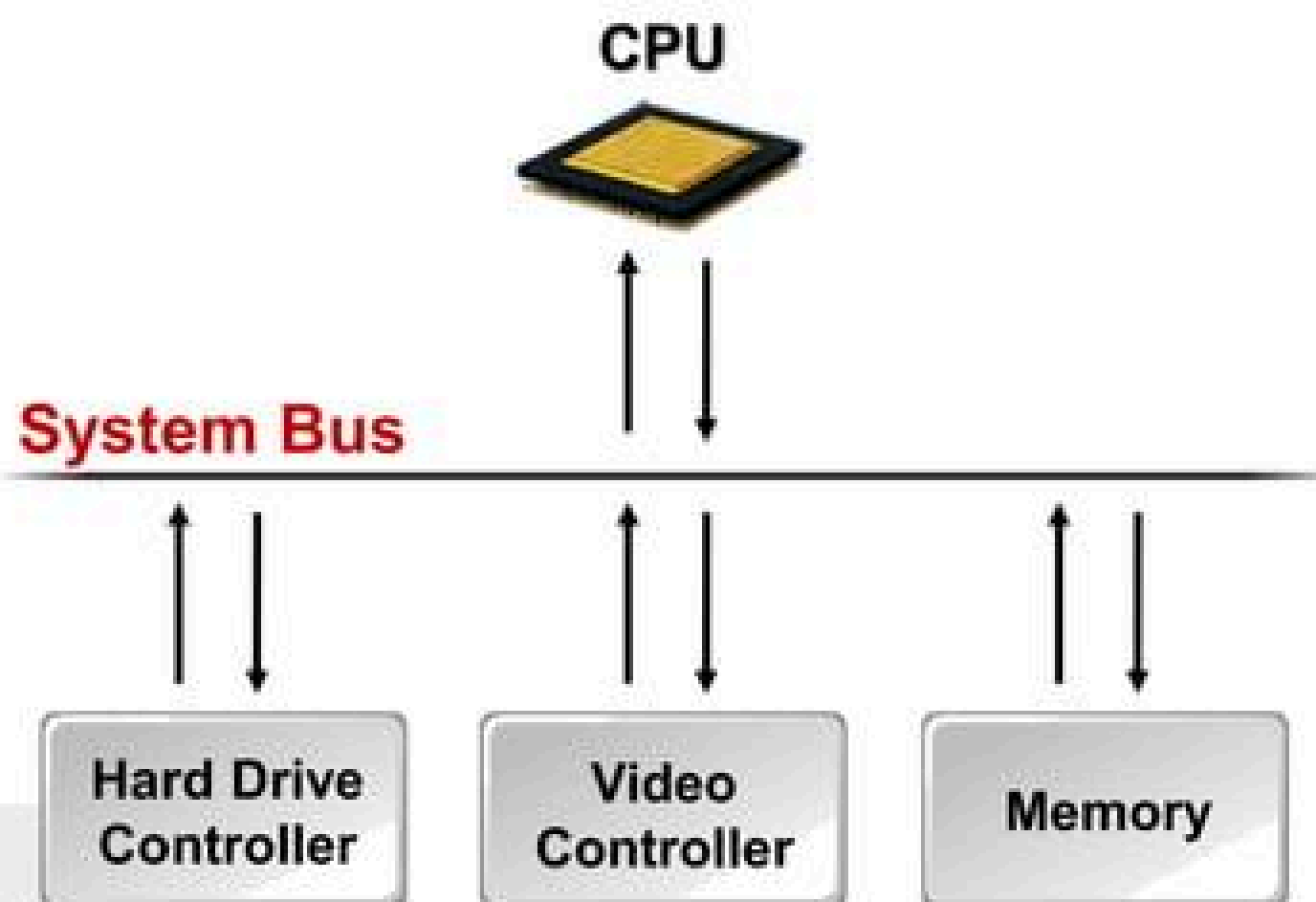
Memory

Memory is the personal computer component that comprises the electronic storage areas in the computer. It can be considered either volatile or non-volatile.



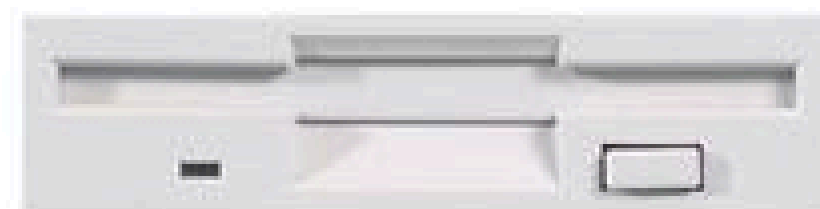
The System Bus

The ***system bus*** is the main communication path between the CPU and memory.



Storage Devices

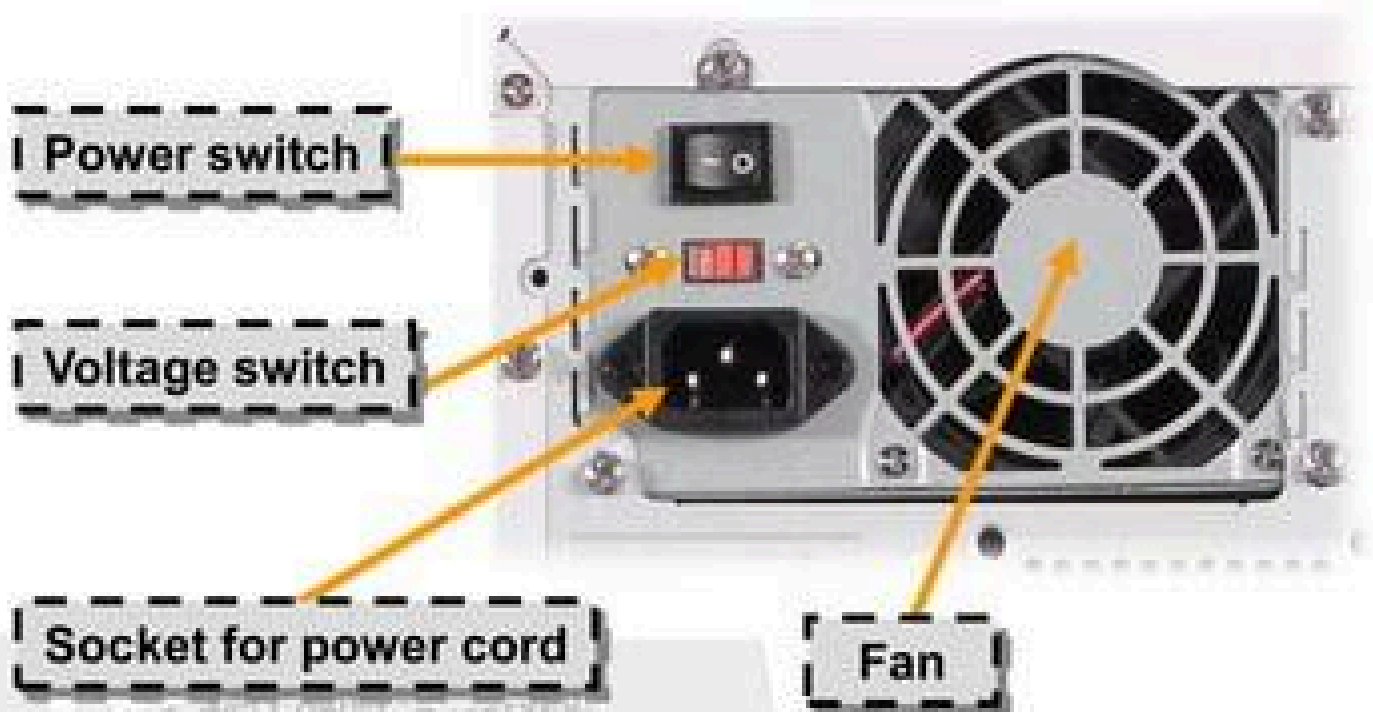
A ***storage device*** is a system unit component, such as a hard drive, that enables users to save data for reuse at a later time, even after the personal computer is shut down and restarted.



Power Supplies

A ***power supply*** is an internal computer component that converts AC power from an electrical outlet to the DC power needed by system components.

External view



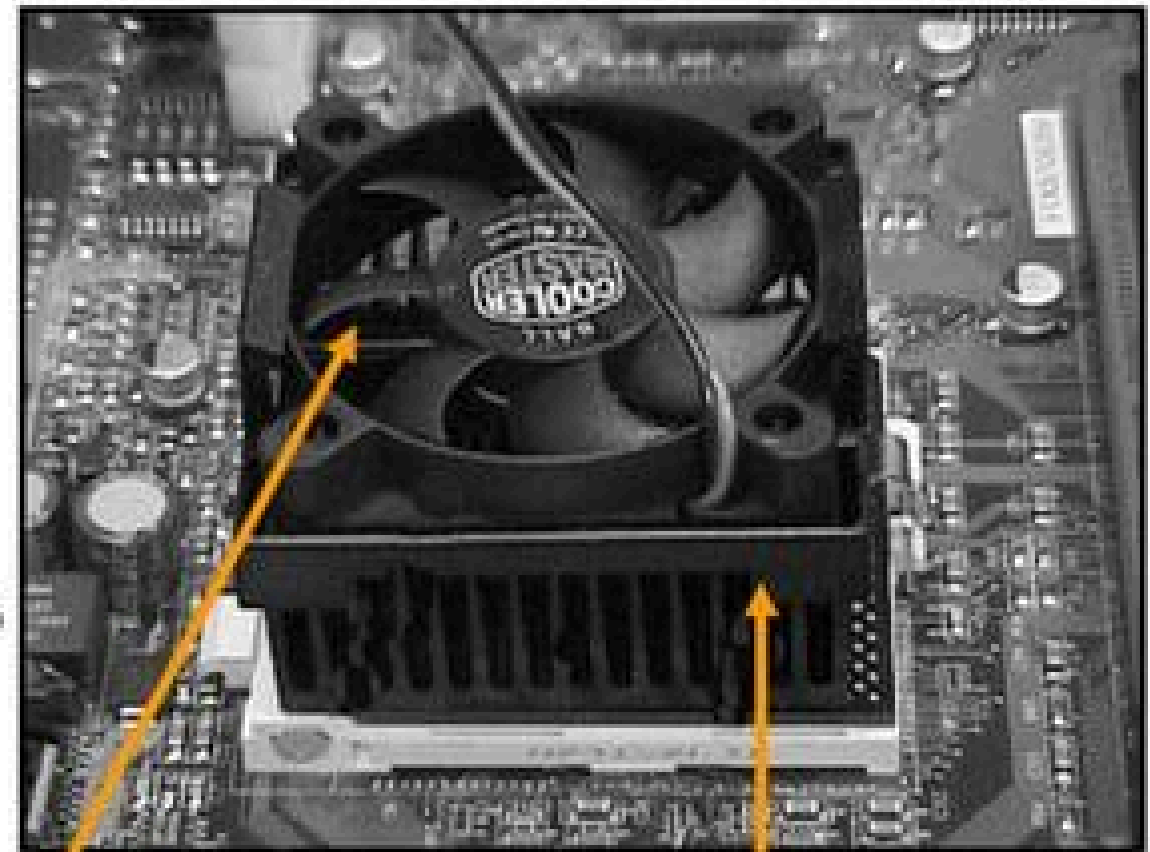
Internal view



Cooling Systems

Computer systems contain several components that require cooling:

- The computer case
- The CPU
- The power supply
- Some adapter cards
- Some hard disk drives

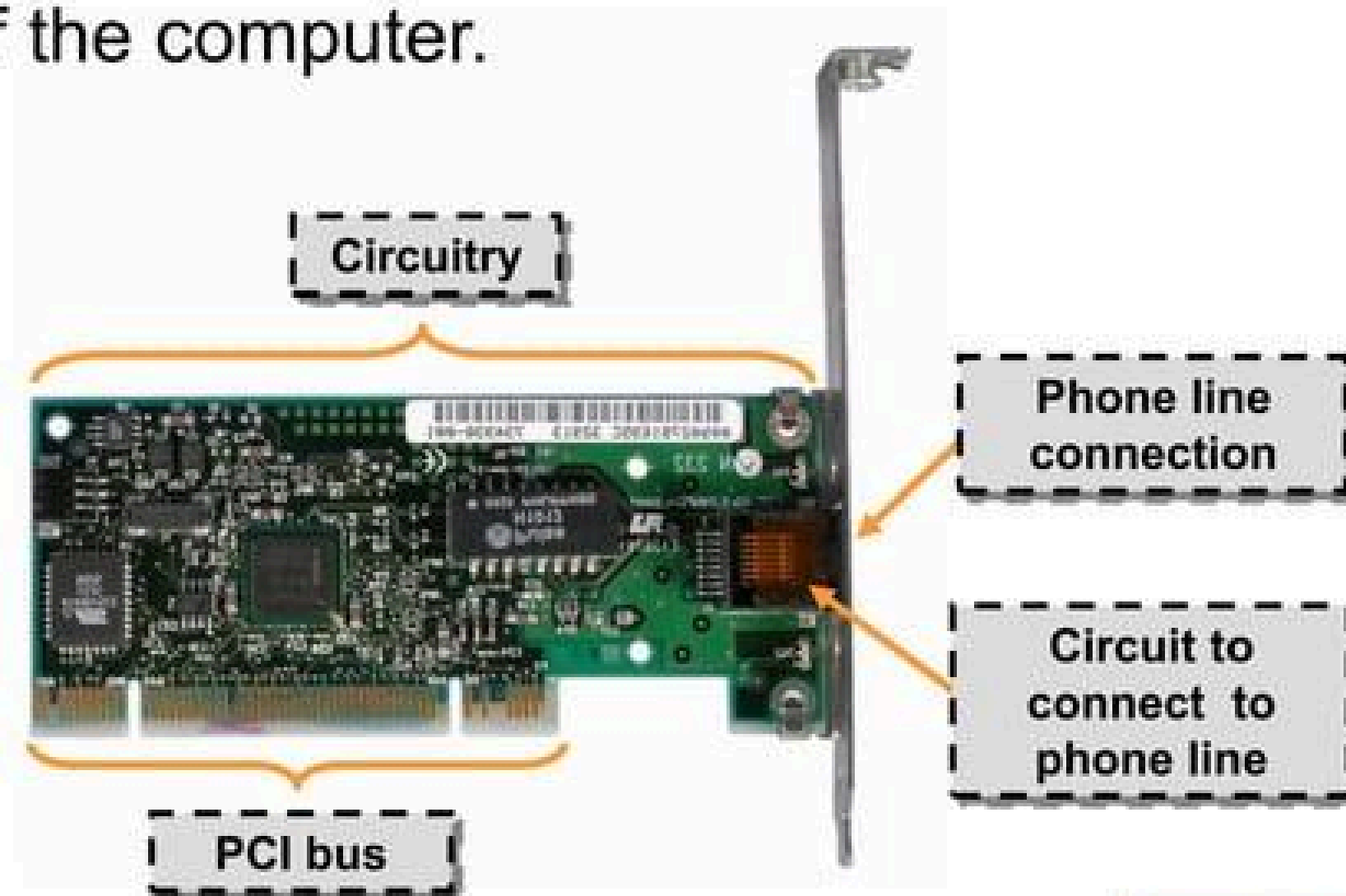


Fan

Heat sink

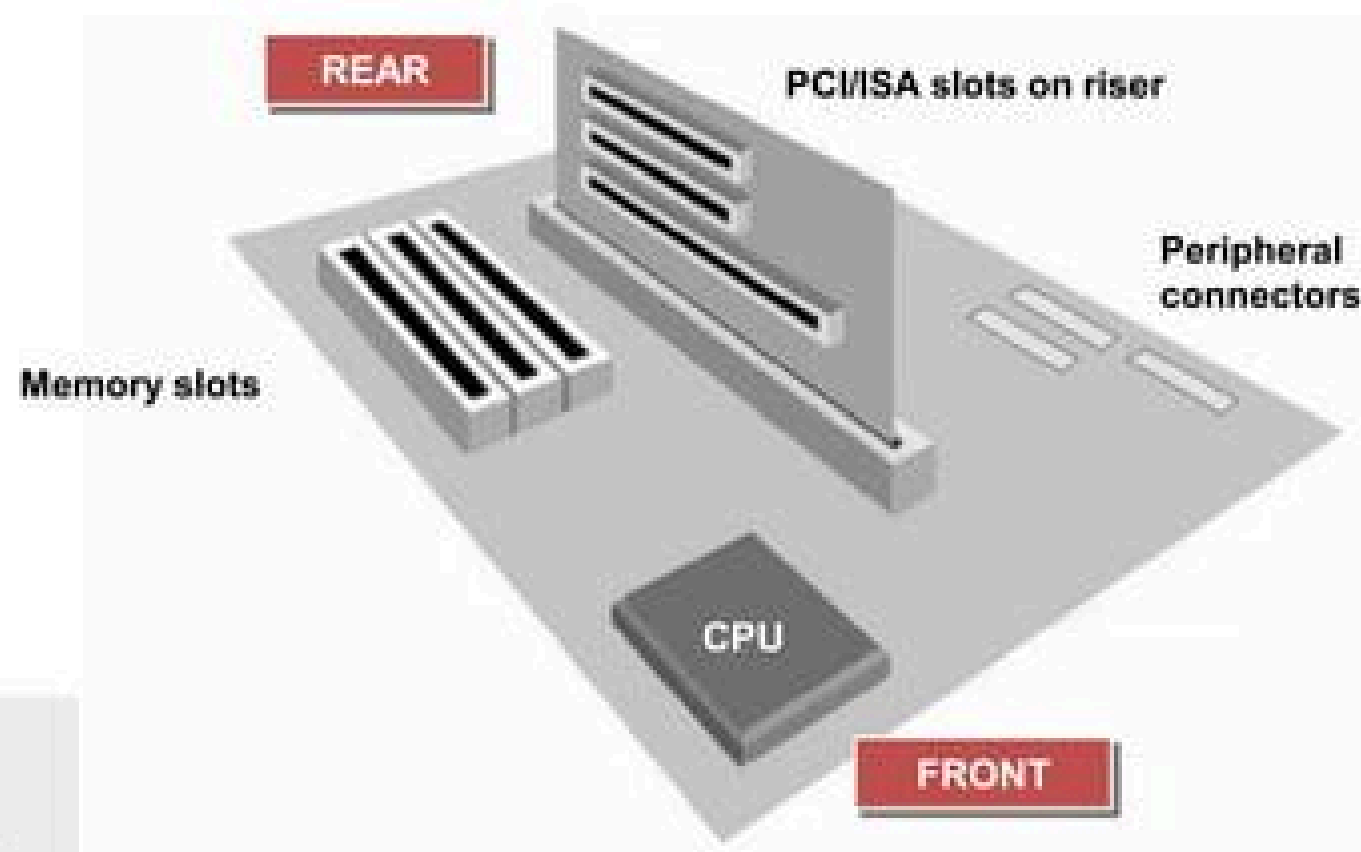
Adapter Cards

An ***adapter card*** is a printed circuit board that you install into a slot on the computer's system board to expand the functionality of the computer.



Riser Cards

A ***riser card*** is a board that plugs in to the system board and provides additional slots for adapter cards..



Storage Devices

The various types of storage devices used in personal computers are:

- ➔ Floppy Disk Drives (FDD)
- ➔ Hard Disk Drives (HDD)
- ➔ Optical Drives
- ➔ Tape Drives
- ➔ Solid State Storage devices

Floppy Disk Drives (FDD)

FDD is a personal computer storage device that reads data from, and writes data to, removable disks made of flexible Mylar plastic covered with a magnetic coating and enclosed in a stiff, protective, plastic case.

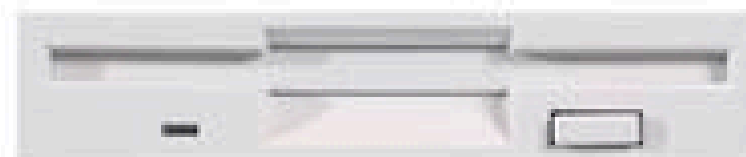
Rear view of floppy disk drive



Data cable

Power cable

Front view 3.5-inch floppy disk drive



3.5-inch floppy disk

Hard Disk Drives (HDD)

HDD is a personal computer storage device that uses fixed media, which means that the disk is built into the drive and the drives are not removed from the computer unless you are performing an upgrade or a repair.



Optical Disks

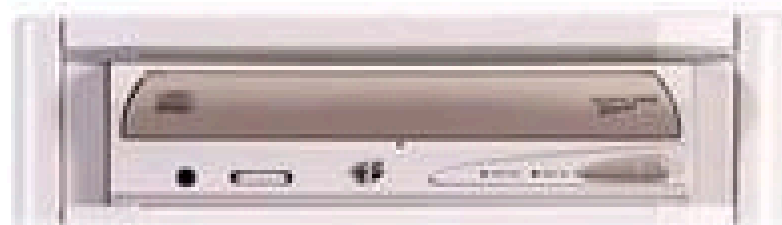
An ***optical disk*** is a personal computer storage device that stores data optically, rather than magnetically.



An external DVD drive



A CD-R disc



An internal CD-ROM drive



Types of Optical Disks and Drives

Optical disks and drives come in several types, such as:

- CD-ROM
- CD-R
- CD-RW
- DVD-ROM
- DVD-R
- DVD+R
- DVD+R DL
- DVD-RW
- DVD+RW
- DVD-RAM



Tape Drives

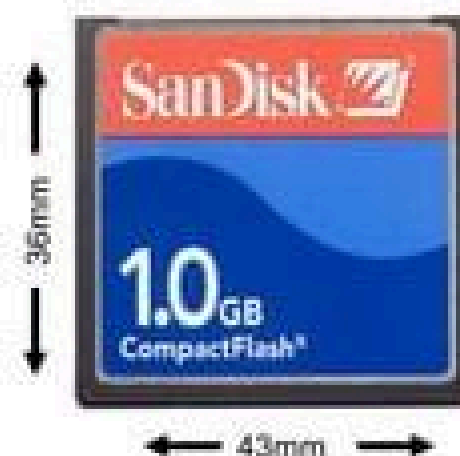
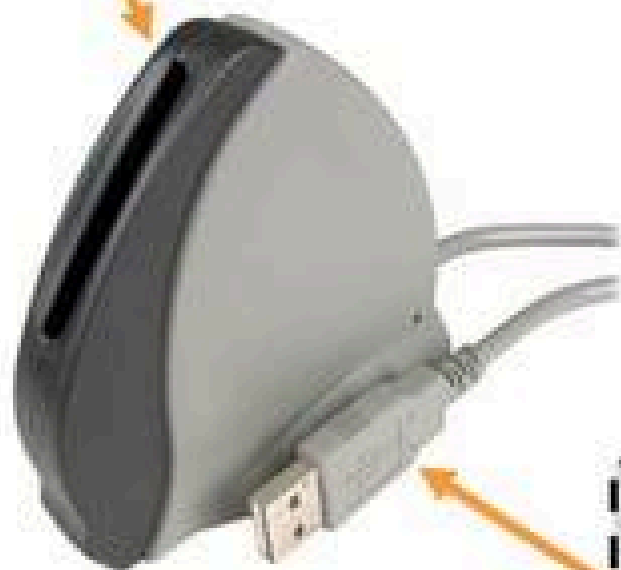
A tape drive is a personal computer storage device that stores data magnetically on a removable tape that is enclosed in a tape cartridge.



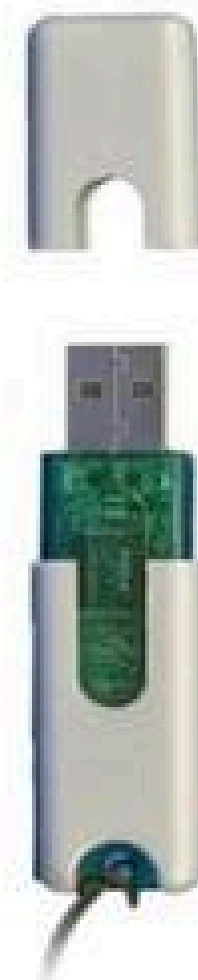
Solid

Solid state storage is a personal computer storage device that stores data in special types of memory instead of on disks or tape.

Compact Flash
card fits in here



USB adapter connects the
Compact Flash reader to
the computer



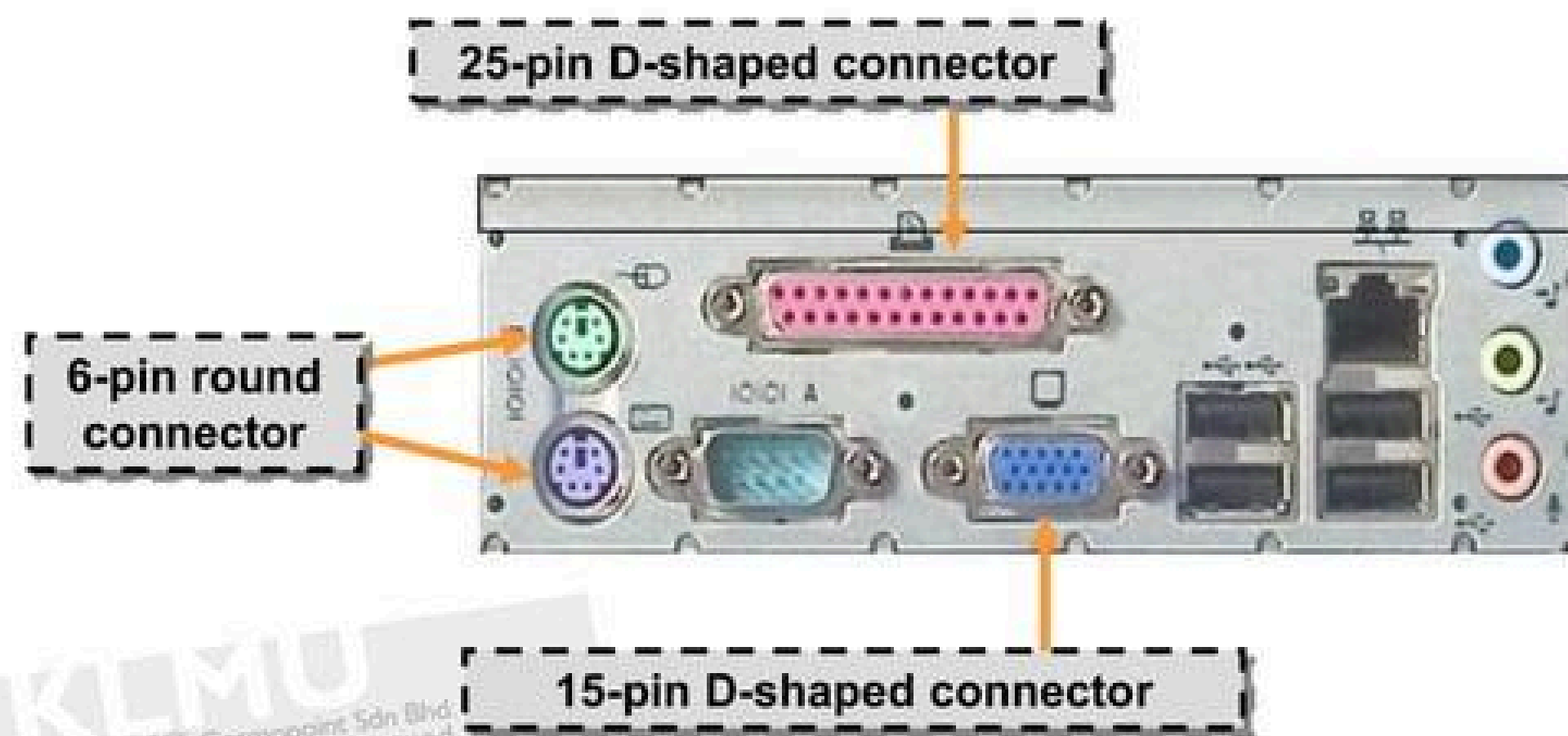
Personal Computer Connection Methods

The various PC connection methods are:

- Ports
- Personal Computer Connections
- Serial Connections
- Parallel Connections
- Universal Serial Bus (USB) Connections
- IEEE 1394 and FireWire Connections
- Small Computer Systems Interface (SCSI) Connections
- Parallel ATA (PATA) Connections
- Serial ATA (SATA) Connections

Ports

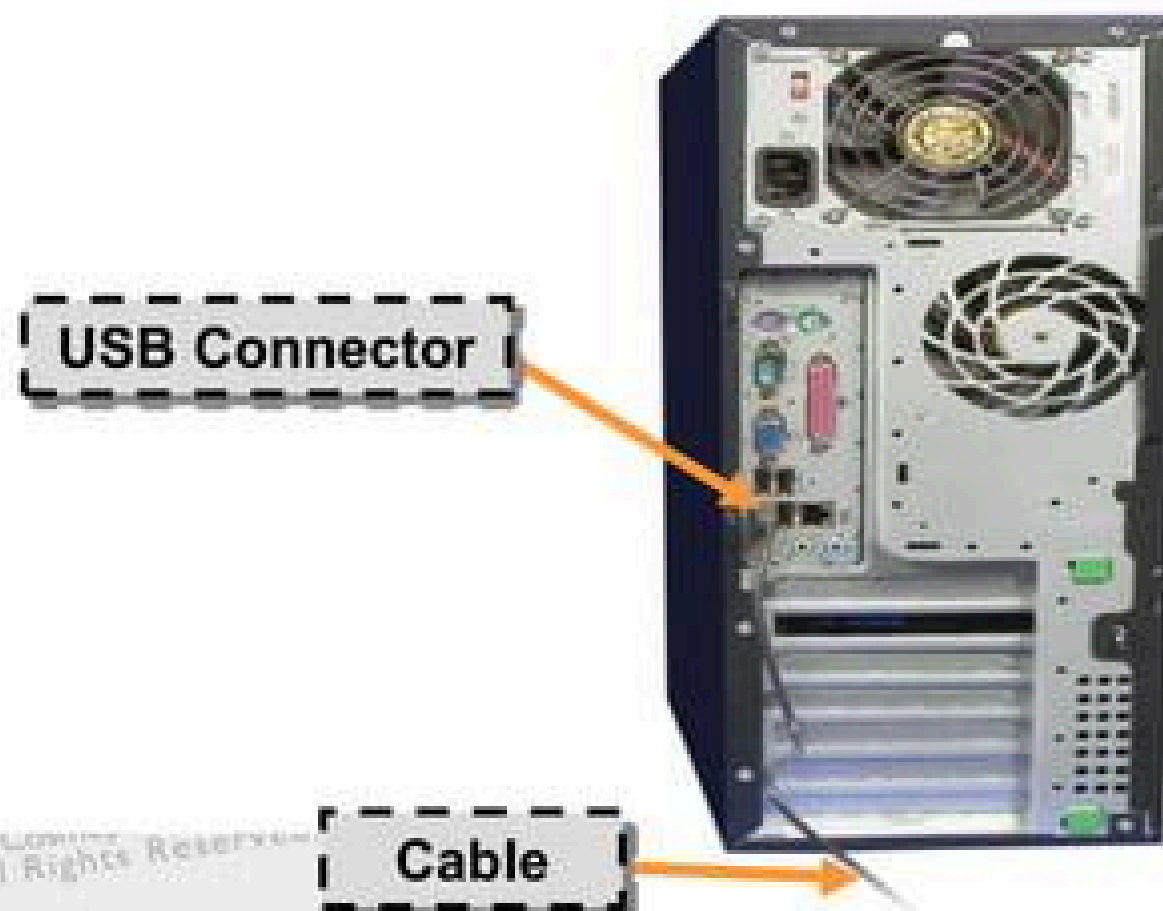
A port is a hardware connection interface on a personal computer that enables devices to be connected to the computer.



Enable devices to be connected to a personal computer.

Personal Computer Connection

A ***personal computer connection*** is a means by which a personal computer component is attached to other components to provide computing capabilities.



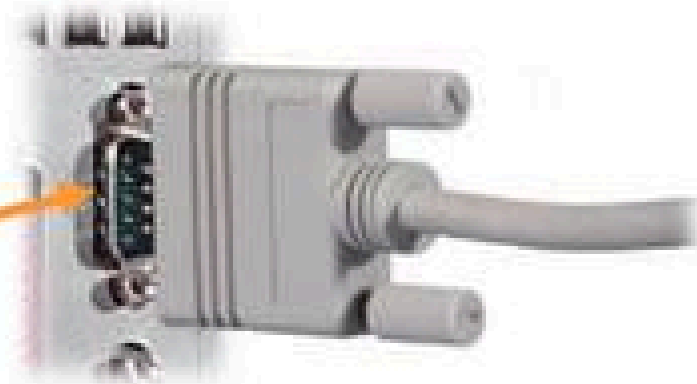
Serial Connections

A ***serial connection*** is a personal computer connection that transfers data one bit at a time over a single wire.

25-pin serial port
on a serial device

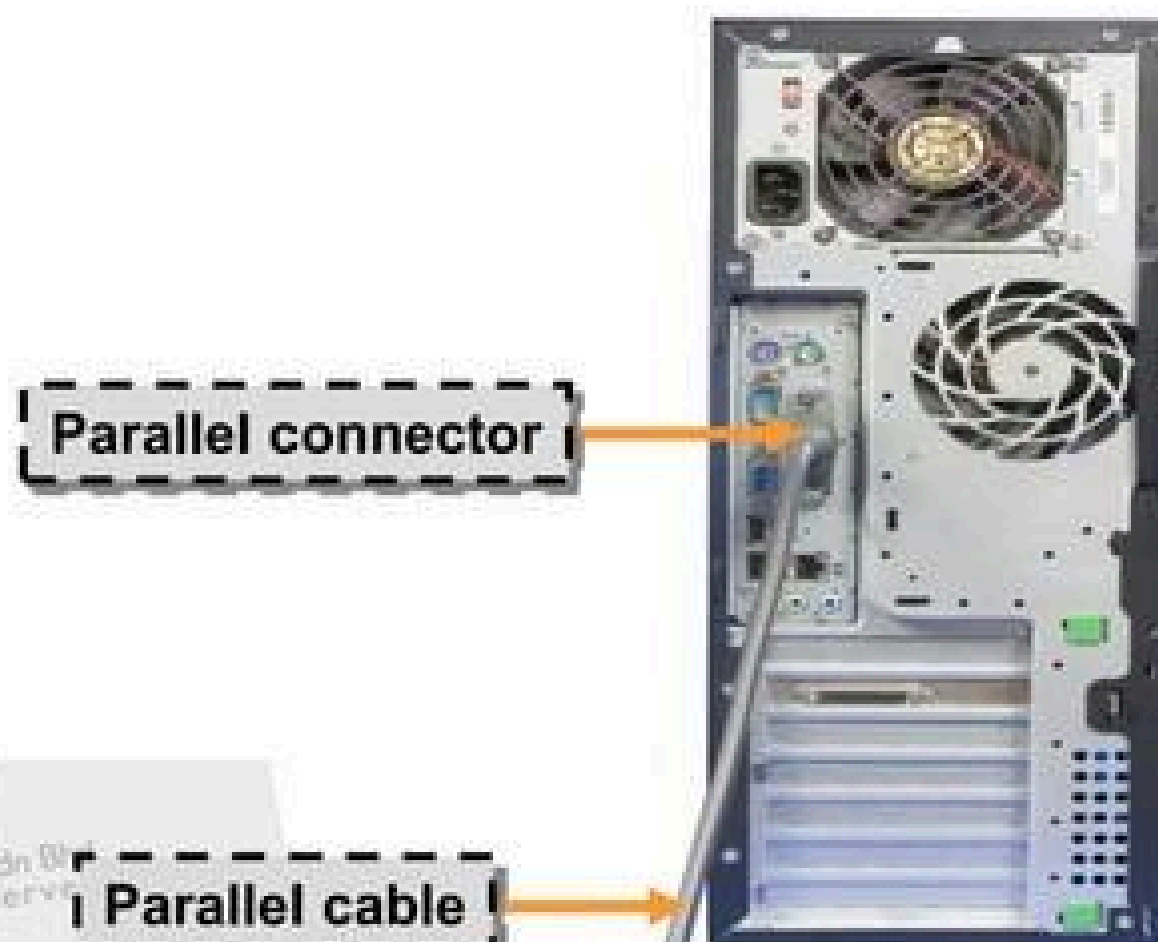


25-pin end of serial cable connects to modem
and 9-pin end connects to computer's serial port



Parallel Connections

A ***parallel connection*** is a personal computer connection that transfers data eight bits at a time over eight wires and is typically used to connect a printer to a system unit.



USB Connections

A **USB connection** is a personal computer connection that enables you to connect multiple peripherals to a single port with high performance and minimal device configuration.



Type B connector connects to USB device

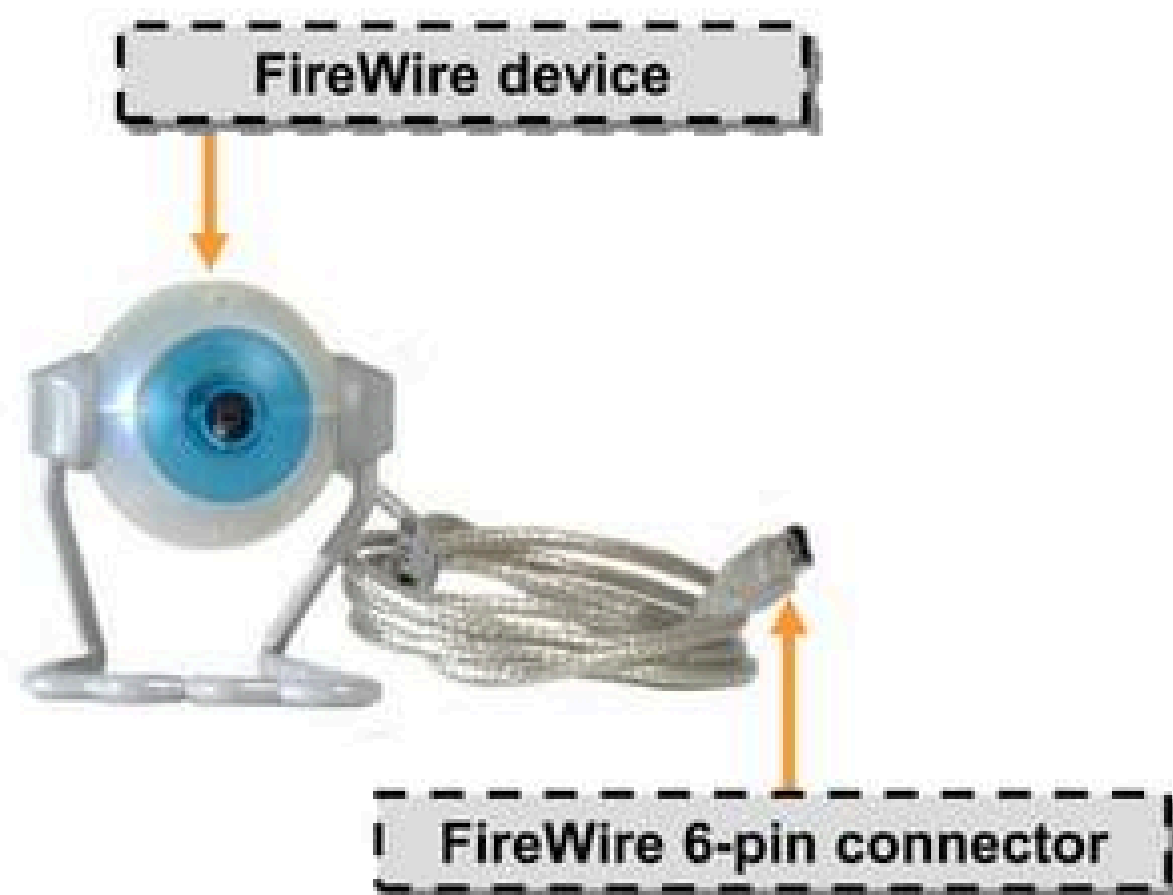
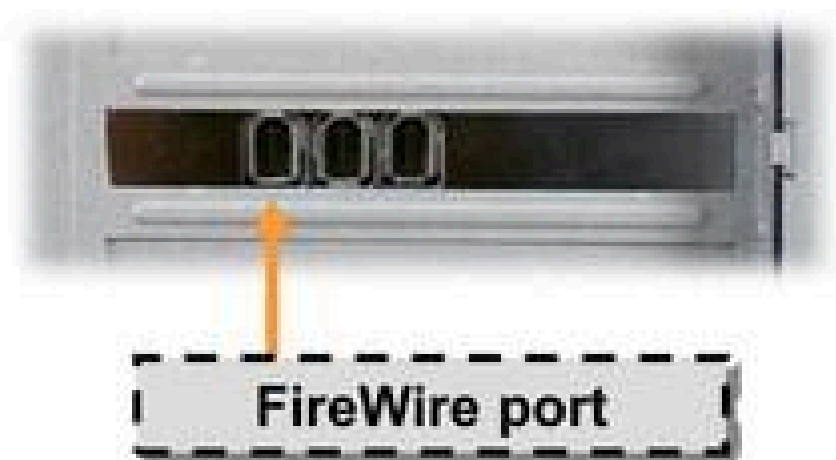


Type A connector connects to USB port on the computer or on a hub



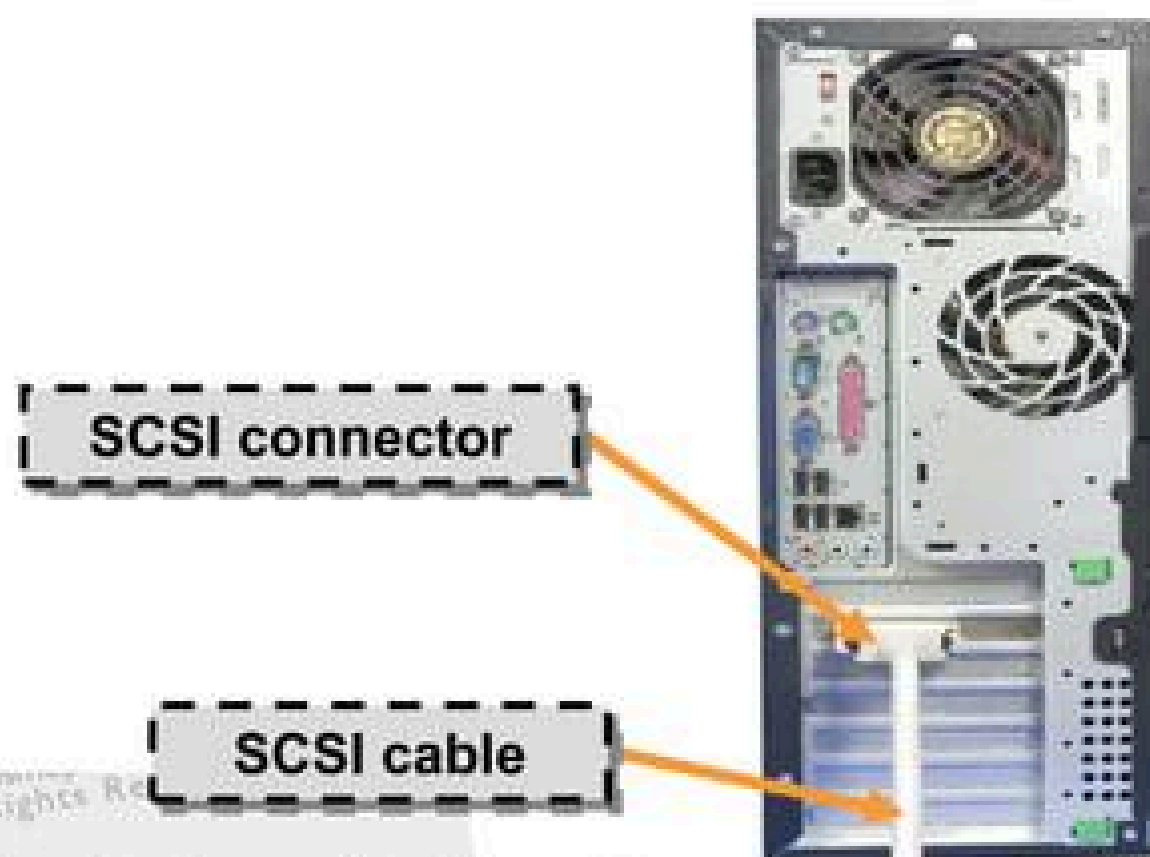
IEEE 1394 and FireWire Connections

A ***FireWire connection*** is a personal computer connection that provides a high-speed interface for peripheral devices that are designed to use the IEEE 1394 standard.



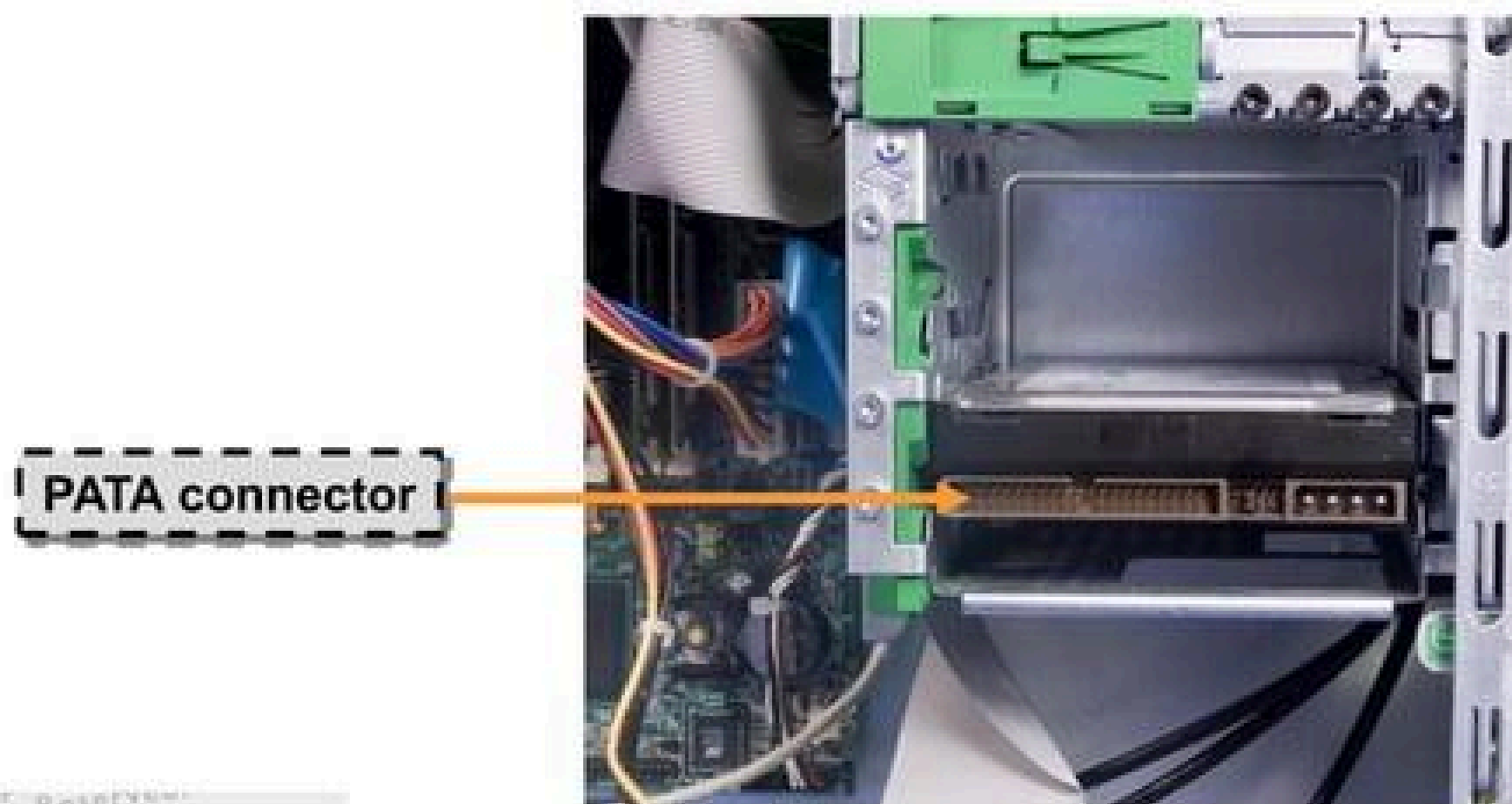
Small Computer Systems Interface (SCSI) Connections

A **SCSI connection** is a personal computer connection that connects internal and external components to the system unit and provides for high-speed data transfer.



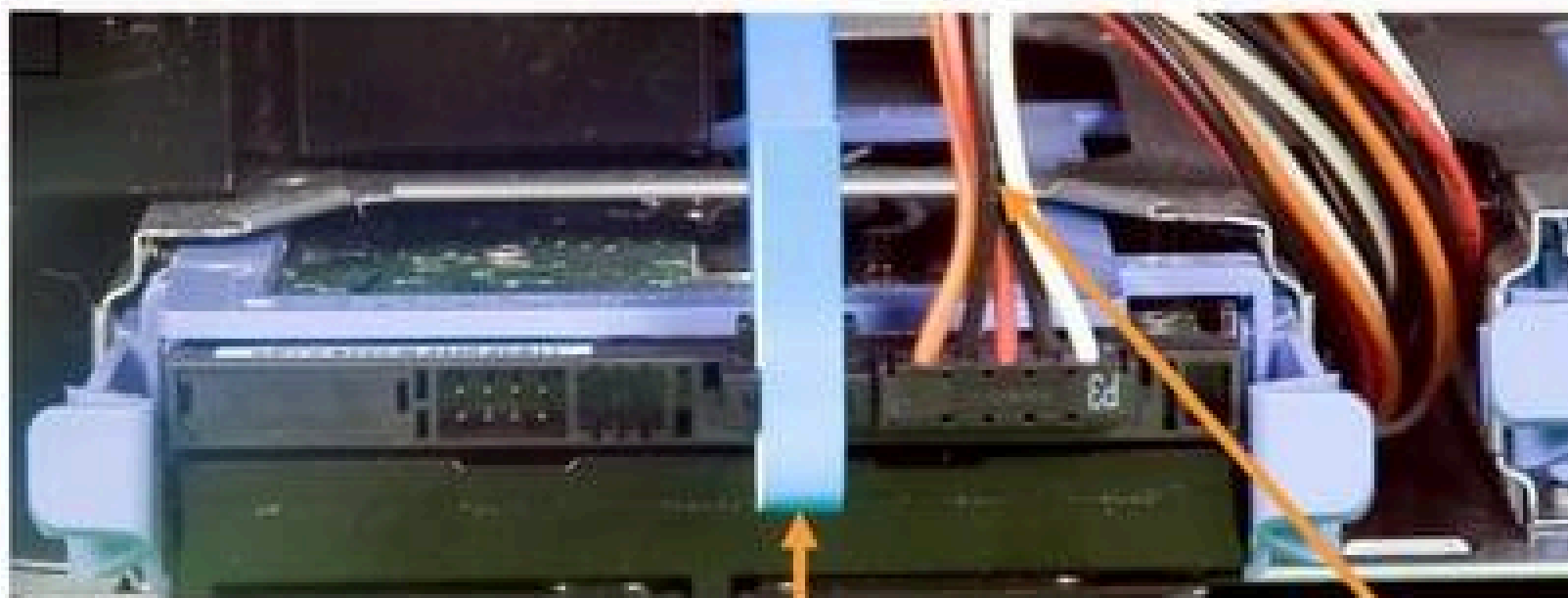
Parallel ATA (PATA) Connections

A ***PATA connection*** is a personal computer connection that provides a parallel data channel from a disk controller to the disk drives.



Serial ATA (SATA) Connections

A ***SATA connection*** is a personal computer connection that provides a serial data channel between the drive controller and the disk drives.



SATA data cable

SATA power cables

1.5 Categories of Computer

TOPIC

Personal
Computer

Mobile Computer
and Mobile
Devices

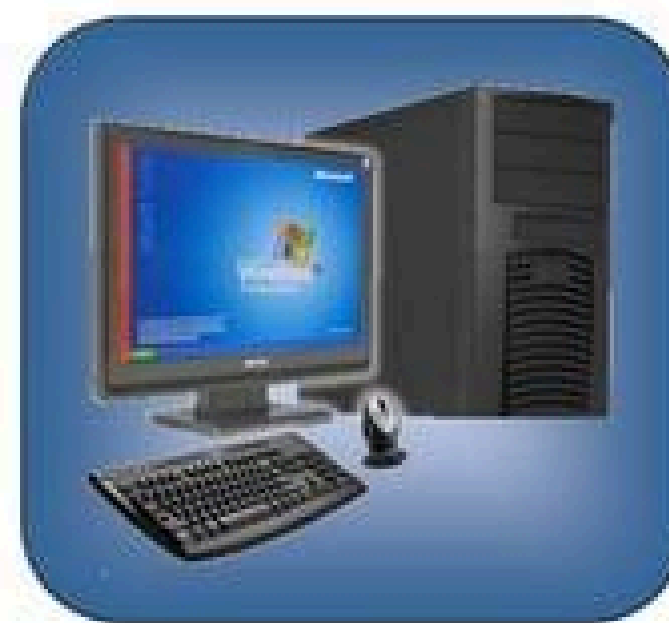
Mid-Range Server

Mainframe
Computer

Supercomputer

Personal Computer

- ❑ A single-user computer that can perform all input, processing, output and storage operations on its own.
- ❑ Two types of personal computer are desktop computer and workstation.



Notebook Computer

- ☐ Portable, small enough to fit on your lap.
- ☐ Also called a **laptop computer**.
- ☐ Usually more expensive than a desktop computer with equal capabilities.



Tablet PC

- ☐ A computer that is shaped like a slate or notebook.
- ☐ Normally fitted with a touch screen for users to enter input through a stylus.



Handheld Computer

- ❑ A small computer that can easily be fit in one hand and operated by the other hand.



Personal Digital Assistant (PDA)

- ❑ A small, portable computer built for specific purposes such as for keeping phone directories and calendars.



Mid-Range Server

- ❑ More powerful and larger than a workstation and can support up to 4,000 users at one time.
- ❑ Formerly known as minicomputer.



Mainframe Computer

- ❑ Very powerful, expensive computer that supports thousands of connected users



Supercomputer

- ☐ The fastest, most powerful, most expensive computer.
- ☐ Used for applications requiring complex mathematical calculations.

